Interrogative and standard disjunction in Mandarin Chinese

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Mandarin Chinese lexically distinguishes the disjunctors in alternative questions (*háishi*) and in disjunctive assertions (*huòzhe*), reflecting a distinction that Haspelmath (2007) and others have called *interrogative* versus *standard disjunction*. I argue that the two disjunctors share their basic syntax and semantics as junction heads (J) that project their disjuncts as Roothian alternatives, which are then interpreted by a corresponding question-forming operator or existential operator. I motivate this view from island insensitivity and focus intervention effects, which I show to apply in parallel to both alternative question formation with *háishi* and the scope-taking of *huòzhe*.

Háishi also allows for a number of non-interrogative uses, subject to significant speaker variation. I argue that these patterns reflect broadly two types of grammars: those where *háishi* syntactically enforces that its alternatives be interpreted by certain operators, and those that do not. For the latter, more liberal speakers, *háishi* can be used non-interrogatively in the same environments that *wh*-phrases can be. The study and analysis of this pattern of variation leads to the conclusion that a so-called "interrogative disjunction" could be so specified via its syntactic specification or through its semantics alone, with both strategies being attested by different sets of Mandarin Chinese speakers.

Keywords interrogative disjunction, alternative questions, non-interrogative *wh*-words, speaker variation, Alternative Semantics, Mandarin Chinese

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Contents

1 Introduction

Mandarin Chinese has two disjunctors: *háishi* and *huòzhe*. Consider the sentences in (1a) and (1b), which are superficially identical except for the choice of disjunction. Example (1a) uses *háishi* and is interpreted as an alternative question, which is answered by identifying which person Zhang San likes. Equivalents of 'yes' or 'no' are not valid replies to (1a) (Li and Thompson, 1981: 558–561). In contrast, (1b) uses *huòzhe* and must be interpreted as a declarative expressing a logical disjunction.[1]

(1) **Interrogative and standard disjunctors in Mandarin Chinese:**

a. <u>*háishi* ⇒ alternative question:</u>

Zhāng Sān xǐhuān Lǐ Sì *háishi* Wáng Wǔ (ne)?
Zhang San like Li Si IDISJ Wang Wu NE
'Does Zhang San like Li Si or Wang Wu?' (alternative question)

b. <u>*huòzhe* ⇒ disjunctive statement:</u>

Zhāng Sān xǐhuān Lǐ Sì *huòzhe* Wáng Wǔ.
Zhang San like Li Si SDISJ Wang Wu
'Zhang San likes Li Si or Wang Wu.'

I will refer to and gloss *háishi* as *interrogative disjunction* (IDISJ) and *huòzhe* as *standard disjunction* (SDISJ). I adopt these terms from Haspelmath (2007) and Mauri (2008) who discuss pairs of this type from a typological perspective.

The first contribution of this paper will be to present a concrete compositional semantics for these two disjunctors. I argue that both disjunctors share their basic syntax and semantics. Syntactically, both are J (junction) heads (Den Dikken, 2006) that take disjuncts of variable size. Semantically, I follow prior approaches that treat disjunction as introducing a set of alternatives (e.g. Aloni, 2003, 2007; Simons, 2005; Alonso-Ovalle, 2006) but recast this intuition within the two-dimensional Alternative Semantics framework of Rooth 1985, 1992 et seq, following Beck and Kim 2006. Both disjunctors project their disjuncts as a set of alternatives, with no defined ordinary value, parallel to the semantics for *wh*-phrases (e.g. Beck, 2006; Kotek, 2019, Erlewine, to appear). In simple cases as in (1a,b) above, alternatives projected by *háishi* will be interpreted by a question operator Q, leading to an alternative

[1] The disjunctor *huòzhe* can also be *huòshì* or simply *huò*, which are generally interchangeable (see e.g. Tsai, 2015a: 60ff). Lü (1980: 196) notes that *huò* is associated more with written texts, which Jing-Schmidt and Peng (2016) confirm through a corpus study (see pp. 108–109 note 7). For uniformity, here I use *huòzhe* throughout.

Note that the standard disjunctor *huòzhe* can also be part of a question, such as as a *wh*-question or a polar question, but these questions differ from an alternative question as in (1a) that seeks to choose between the different disjuncts. See note 8.

question interpretation, whereas those projected by *huòzhe* will be interpreted by an existential operator ∃ that produces a disjunctive proposition. This correspondence between the forms of disjunctors and their corresponding operators will be enforced partially but not entirely by syntactic feature-checking, as I discuss further below.

I present my core proposal and necessary theoretical background in section 2. I then present motivating evidence for my proposal from parallels between the two types of disjunctions in their insensitivity to islands and susceptibility to focus intervention effects (as in Beck, 2006; Beck and Kim, 2006), in section 3. This evidence is supplemented by arguments for the proposed syntax of disjunction in Appendix A, in particular arguing against prior approaches that treat *háishi* interrogative disjunction as uniformly taking disjuncts of underlyingly clausal size.

The second contribution of this paper will be to address and account for the distribution of non-interrogative uses of *háishi*, in section 4. For example, *háishi* can be interpreted universally with the quantificational particle *dōu* as in (2a) and as a wide-scope existential that reflects speaker ignorance and irrelevance as in (2b). Additionally, for a subset of speakers, *háishi* allows for narrow-scope existential uses in various licensing environments, such as negation in (2c). The mark % on (2c) reflects this variation in judgments.

(2) **Non-interrogative uses of *háishi*:**

a. Universal *háishi* with *dōu*: (Huang, Li, and Li, 2009: 242)

Júzi *háishi* píngguǒ dōu xíng.
orange IDISJ apple DOU okay
'Oranges and apples are both ok.'

b. Existential *háishi* with speaker ignorance/irrelevance:

Zhāng Sān mǎi-le píngguǒ *háishi* lízi, dàn wǒ bù zhīdào shì nǎ-yī-ge.
Zhang San buy-PFV apple IDISJ pear but 1sg NEG know SHI which-one-CL
'Zhang San bought apples or pears, but I don't know which one.'

c. Existential *háishi* under negation, for some speakers: (Hsieh, 2004: 89)

%Wǒ méiyǒu kànjiàn Zhāng Sān *háishi* Lǐ Sì.
1sg NEG.PFV see Zhang San IDISJ Li Si
'I didn't see Zhang San nor Li Si.'

Based on patterns of judgments reported in prior literature (especially in Lin Hsin-yin 2008 and Yuan Mengxi 2021) as well as that of speakers that I have consulted, I will show that there are broadly two groups of Mandarin speakers. For the first group with a more restricted distribution of non-interrogative *háishi*, I will argue that *háishi* must be syntactically associated with an operator in a high, clause-typing domain of the clause, such as a question operator Q (which I argue to also be present in *háishi...dōu* structures such as in (2a)) or an operator which derives existential assertions with ignorance inferences. For the other group of speakers who allow non-interrogative *háishi* with narrow-scope existential interpretation in various contexts, such as in (2c), I propose that *háishi* does *not* syntactically enforce its conventional cooccurrence with a particular interpreting operator. *Háishi* disjunctions are then syntactically and semantically equivalent to that of *wh*-phrases for these speakers, simply with a domain corresponding to the named disjuncts. We then correctly predict that, for the second group of speakers, *háishi* can be used non-interrogatively in the same environments where *wh*-phrases can be, as has been observed previously by Lin Hsin-yin (2008). For all speakers, I propose that the standard disjunctor *huòzhe* syntactically enforces its interpretation by a corresponding existential operator, blocking its alternatives from being used to form an alternative question.

My proposal for the attested variation in these non-interrogative uses of *háishi* thus speaks directly to the question of what it means for *háishi* to be an "interrogative disjunction." That is, we could imagine the link between an interrogative disjunction and alternative questionhood being enforced through syntactic mechanisms or merely through the compositional semantics. My proposal here demonstrates that both strategies are possible and indeed attested in language; the first group of Mandarin speakers enforces this link syntactically, while the second group does not. This in turn suggests lessons for the analysis of so-called interrogative disjunctions in other languages as well, which I conclude with in section 5.

2 Proposal

My proposal is couched within Alternative Semantics (Rooth, 1985, 1992) and its extension to interrogatives, which builds on Hamblin 1973 and is developed further in Beck 2006, Kotek 2019, and Erlewine to appear. The key features of this unified framework, which I call *Rooth-Hamblin Alternative Semantics*, is that it is two-dimensional and that the same alternative set dimension is used both for the computation of (Roothian) focus alternatives and (Hamblinian) interrogative alternatives.

In Alternative Semantics, each node α in the syntax is associated with two meanings in different "dimensions": the ordinary semantic value $[\![\alpha]\!]^{\mathrm{o}}$ and a set of alternatives $[\![\alpha]\!]^{\mathrm{alt}}$, with the latter corresponding to the "focus-semantic value" $[\![\alpha]\!]^{\mathrm{f}}$ of Rooth 1992. The interpreted meaning of an utterance is its ordinary semantic value. Alternative sets are computed compositionally parallel to the computation of ordinary semantic values, using a process of *pointwise composition*.

I propose that full interpreted structures must satisfy the constraint I call *Interpretability* in (3) below. This Interpretability filter in (3) synthesizes requirements that have been independently proposed as the "Principle of Interpretability" by Beck (2006: 16) (requiring the ordinary value to be defined) and as part of the "Focus Interpretation Principle" by Rooth (1992: 90) (requiring the ordinary value to be a member of the alternative set).

(3) **Interpretability:** (Erlewine 2019, in prep.)
To interpret α, $[\![\alpha]\!]^{\mathrm{o}}$ must be defined and $\in [\![\alpha]\!]^{\mathrm{alt}}$.

By default, the alternative set for a node α is simply the singleton set with its ordinary value, $\{[\![\alpha]\!]^{\mathrm{o}}\}$, thus trivially satisfying Interpretability. However, certain types of structures will have meanings that violate Interpretability, then necessitating some form of "repair" before Interpretability is checked for the full structure.[2] In this way, the enforcement of Interpretability over full interpreted structures will play a key role in motivating the application of particular semantic operations in my proposal below.

I first describe the function of J, the abstract, polyadic functional head underlying disjunctions (Den Dikken, 2006), which is the common core of both *háishi* and *huòzhe*. Following von Stechow 1991 (p. 53), JP's alternative set denotation is the union of its disjuncts' alternative set denotations. The ordinary semantic value of JP is undefined.

(4) **The semantics of J:**

a. $[\![\mathrm{J}\ x_1, ..., x_n]\!]^{\mathrm{o}}$ undefined

b. $[\![\mathrm{J}\ x_1, ..., x_n]\!]^{\mathrm{alt}} = [\![x_1]\!]^{\mathrm{alt}} \cup ... \cup [\![x_n]\!]^{\mathrm{alt}}$

J here is defined for an arbitrary number of arguments, although in most examples here I will illustrate its use with two disjuncts. For example, consider the disjunction of two NPs of type *e*, Li Si and Wang Wu, as in the examples in (1a) above. I assume each disjunct to have a singleton alternative set denotation[3]: $[\![$Lǐ Sì$]\!]^{\mathrm{alt}}$ = {Li Si} and $[\![$Wáng Wǔ$]\!]^{\mathrm{alt}}$ = {Wang Wu}, resulting in $[\![[_{\mathrm{JP}}$ LS or$_{\mathrm{J}}$ WW$]]\!]^{\mathrm{alt}}$ = {Li Si} ∪ {Wang Wu} = {Li Si, Wang Wu} (5b).

(5) a. $[\![\ [_{\mathrm{JP}}\ [_{\mathrm{NP}}\ \text{Lǐ Sì}]\ \mathrm{J}\ [_{\mathrm{NP}}\ \text{W. Wǔ}]]\ ]\!]^{\mathrm{o}}$ undefined b. $[\![\ [_{\mathrm{JP}}\ [_{\mathrm{NP}}\ \text{Lǐ Sì}]\ \mathrm{J}\ [_{\mathrm{NP}}\ \text{W. Wǔ}]]\ ]\!]^{\mathrm{alt}} = \{\text{Li Si, Wang Wu}\}$

[2] Interpretability may be a more general requirement of all linguistic units of a particular size, such as every full clause (Erlewine 2019) or phase, but its enforcement at over full utterances suffices here.

As a reviewer notes, both *háishi* and *huòzhe* disjunctors can appear more than once in a disjunction of three or more disjuncts. Example (6) illustrates this for the interrogative disjunctor *háishi*.[4] The simple set disjunction semantics for J in (4) extends to these cases as well, as illustrated by the tree in (7) below, which reflects the complex disjunction in example (6) below, with alternative set denotations indicated for each node.

(6) **Disjunctions with multiple disjunctors:** (Li and Thompson, 1981: 535)

Tā	jīntiān	(*háishi*)	míngtiān	*háishi*	hòutiān	lái?
3sg	today	IDISJ	tomorrow	IDISJ	day.after.tomorrow	come

'Is he/she coming today, tomorrow, or the day after?'

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[3] To say that each disjunct has a singleton alternative set denotation is to say that they do not project non-trivial focus alternatives, which may be satisfied by disjuncts not being focused. Disjuncts often do bear focus, but these are cases of contrasting phrases, akin to Rooth's (1992) *American farmer* versus *Canadian farmer* cases. Following Rooth's discussion there, I assume that surface structures with focused disjuncts of the form "$[\text{LS}]_{\text{F}}$ *háishi/huòzhe* $[\text{WW}]_{\text{F}}$" in fact have LF structures of the form $[_{\text{JP}}\ [\sim [\text{LS}]_{\text{F}}]\ \text{or}_{\text{J}}\ [\sim [\text{WW}]_{\text{F}}]]$, where $\sim$ is Rooth's "resetting" focus interpretation operator. (I discuss this notion of "resetting" in section 2.2 below.) Similarly, individual disjuncts can include foci as well as corresponding overt focus-sensitive operators, as in example (31) below.

Allowing focus-marking within disjuncts that are not "reset" disjunct-internally would lead to incorrect interpretations. For example, consider the structure $[_{\text{JP}}\ [\text{LS}]_{\text{F}}\ \text{J}\ [\text{WW}]]$ with F-marking on the first disjunct. The denotation for J in (4) then predicts JP to include other, unnamed individuals in its alternative set — $[\![[_{\text{JP}}\ [\text{LS}]_{\text{F}}\ \text{J}\ [\text{WW}]]]\!]^{\text{alt}} = [\![[\text{LS}]_{\text{F}}]\!]^{\text{alt}} \cup [\![\text{WW}]\!]^{\text{alt}} = \{\text{ZS, LS, WW}, \ldots\}$ — because the focused left disjunct will include other alternatives to Li Si besides Wang Wu. This is an unwelcome result.

However, such structures with focus-marking within a disjunct that is not "reset" disjunct-internally is independently ruled out based on an independently motivated constraint of the Rooth-Hamblin Alternative Semantics framework in (i), discussed in Kotek 2019: 46 and Erlewine to appear.

(i) An LF φ containing a focused expression A is infelicitous if, for some B distinct from A, φ is truth-conditionally equivalent to $\varphi[A \to B]$ (the result of substituting B for A in φ).

Concretely, we predict that $[_{\text{JP}}\ [\text{LS}]_{\text{F}}\ \text{or}_{\text{J}}\ [\text{WW}]]$ has the same interpretation as $[_{\text{JP}}\ [\text{ZS}]_{\text{F}}\ \text{or}_{\text{J}}\ [\text{WW}]]$, where we replace the focused constituent Li Si with one of its alternatives, Zhang San. Consequentially, structures of this form will be ruled out by (i).

[4] The same sort of structure is possible with the standard disjunctor *huòzhe* as well, but disjunctors cannot be mixed within a single complex disjunction. I propose that this restriction is enforced syntactically, either via selectional restriction (JP sisters of *háishi* must themselves be headed by *háishi*, and mutatis mutandis for *huòzhe*) or as a variety of agreement or concord.

(7)

JP

{today, tomorrow, the day after}

- NP {today} — *jīntiān*
- J
- JP {tomorrow, the day after}
  - NP {tomorrow} — *míngtiān*
  - J
  - NP {the day after} — *hòutiān*

Alternatives introduced by J will compose *pointwise* with other material in the clause. The tree in (9) illustrates the VP in (8) with the subject in its VP-internal position[5] — following the predicate-internal subject hypothesis for Mandarin (see e.g. Huang, 1993; Shyu, 1995; Lin, 1998a) — with alternative sets and the types of their elements indicated at each node. I give extensional formulations here for ease of presentation.

(8) Zhāng Sān xǐhuān Lǐ Sì *háishi*/*huòzhe* Wáng Wǔ
Zhang San like Li Si IDISJ/SDISJ Wang Wu
háishi: 'Does Zhang San like Li Si or Wang Wu?' (alternative question)
huòzhe: 'Zhang San likes Li Si or Wang Wu.' (declarative)

(9)

VP_t

$\left\{ \begin{array}{l} \textbf{like}(\text{ZS}, \text{LS}), \\ \textbf{like}(\text{ZS}, \text{WW}) \end{array} \right\}$

- NP_e {Zhang San}
- $\text{V'}_{\langle e,t \rangle}$ $\left\{ \begin{array}{l} \lambda y \,.\, \textbf{like}(y, \text{LS}), \\ \lambda y \,.\, \textbf{like}(y, \text{WW}) \end{array} \right\}$
  - $\text{V}_{\langle e,\langle e,t \rangle\rangle}$ $\{\lambda x \,.\, \lambda y \,.\, \textbf{like}(y, x)\}$
  - JP_e {Li Si, Wang Wu}

[5] I use the label VP here as shorthand for the full predicative domain with all of its predicate-internal arguments saturated, which is often called *v*P, VoiceP, or PredP in contemporary syntactic work. Following the basic compositional model as in Heim and Kratzer 1998, without events, what I call VP is always the lowest node in the clausal spine with extensional type *t*.

As reflected in (9), the alternative set for a branching node with daughters β and γ is computed by crossing each denotation in $[\![\beta]\!]^{\text{alt}}$ with each denotation in $[\![\gamma]\!]^{\text{alt}}$ and composing them using the appropriate rule of composition, e.g. function application. Each alternative in $[\![\text{JP}]\!]^{\text{alt}}$ of type e thus corresponds to an alternative of propositional type in $[\![\text{VP}]\!]^{\text{alt}}$. The subject subsequently moves to its canonical subject position in Spec,TP, not illustrated here.

Many previous works share the idea that disjunction collects a set of alternatives that then lead to the computation of corresponding alternatives at higher levels of structure via pointwise composition (Aloni, 2003, 2007; Simons, 2005; Alonso-Ovalle, 2006, 2008, 2009, and more recently Biezma and Rawlins, 2012; Szabolcsi, 2013, 2015; Slade, 2019; cf. also Winter, 1995, 1998). However, these previous proposals are couched in a one-dimensional Hamblin semantics or equivalent. My proposal for J in (4) is a particular implementation of this idea within Rooth's two-dimensional Alternative Semantics.

Note that the node JP in (9) does not have an ordinary semantic value, as defined in (4). The nodes V' and VP, which are dependent on the denotation of JP, will therefore also have undefined ordinary values. This lack of an ordinary semantic value will need to be addressed by the end of the derivation, in order to satisfy Interpretability (3). Some higher alternative-sensitive operator must construct an ordinary semantic value based on the alternatives, so that the utterance root can be interpreted. This will happen in one of two ways: either an existential operator quantifies over these alternatives, resulting in their boolean disjunction, or the alternatives are used to form a question, with each alternative corresponding to a possible answer.

2.1 Alternative questions with *háishi*

We first discuss the use of *háishi*, which in the basic case reflects the use of J (4) above without further modification. Consider the basic alternative question example (1a) from above, repeated here as (10).

(10) **Alternative question with *háishi*:** =(1a)

[$_{\text{TP}}$	Zhāng Sān	xǐhuān	[$_{\text{JP}}$	Lǐ Sì	*háishi*	Wáng Wǔ]]	(ne)?
	Zhang San	like		Li Si	IDISJ	Wang Wu	NE

'Does Zhang San like Li Si or Wang Wu?'

The semantic denotation for the TP clause in (10), modulo the contribution of tense/aspect semantics which I do not consider here, is equal to the denotation of the VP illustrated in (9) above. The alternative set contains two propositions, which I intensionalize here, corresponding to Zhang San liking Li Si and Zhang San liking Wang Wu. Its ordinary semantic value is undefined:

(11) a. $[\![\text{TP}]\!]^{\text{o}}$ undefined b. $[\![\text{TP}]\!]^{\text{alt}} = \{{}^\wedge\textbf{like}(\text{ZS}, \text{LS}),\ {}^\wedge\textbf{like}(\text{ZS}, \text{WW})\}$

The sentence-final particle *ne* is often included in Mandarin alternative questions, as in (10) above. I discuss the behavior of *ne* in Appendix A, but for our current discussion, it suffices to note that this sentence-final particle *ne* is independent of the utterance's interrogative force and is part of the CP layer of the clause (see e.g. Constant, 2014; Paul, 2014). I will then simply treat the full CP as semantically identical to the TP: $[\![\text{CP}]\!] = [\![\text{TP}]\!]$.

Recall that a complete utterance is interpreted as its ordinary semantic value, which must also be a member of its alternative set denotation. The TP/CP here fails to satisfy this requirement of Interpretability (3). We need an operator that defines an ordinary semantic value—in this case of a question—based on the denotation in (11). Beck proposes a question operator Q for this purpose, with the syncategorematic entry in (12) (also articulated earlier in Hong 1995: 57–61). As discussed by Kotek (2019), Q (her AltShift) cannot apply to a sister that already has a defined ordinary value.

(12) **Beck's question operator Q:** (from Beck, 2006: 16; also called AltShift in Kotek 2019: 32)

a. $[\![\text{Q}\ \alpha]\!]^{\text{o}} = [\![\alpha]\!]^{\text{alt}}$

b. $[\![\text{Q}\ \alpha]\!]^{\text{alt}} = \{[\![\text{Q}\ \alpha]\!]^{\text{o}}\} = \left\{[\![\alpha]\!]^{\text{alt}}\right\}$

c. [Q α] presupposes that $[\![\alpha]\!]^{\text{o}}$ is undefined.

Its application to the CP built from (11) results in a set of propositions as its ordinary semantic value, each corresponding to different possible answers, i.e. a question denotation (Hamblin, 1973):

(13) a. $[\![\text{Q CP}]\!]^{\text{o}} = \{{}^\wedge\textbf{like}(\text{ZS}, \text{LS}),\ {}^\wedge\textbf{like}(\text{ZS}, \text{WW})\}$

b. $[\![\text{Q CP}]\!]^{\text{alt}} = \{\{{}^\wedge\textbf{like}(\text{ZS}, \text{LS}),\ {}^\wedge\textbf{like}(\text{ZS}, \text{WW})\}\}$

The idea that an operator, Q, "lifts" a set from the alternative dimension into the ordinary dimension is a foundational part of the compositional semantics of *wh*-words and *wh*-questions in Rooth-Hamblin Alternative Semantics, developed in Beck 2006 and Kotek 2014, 2016, 2019. In these works, *wh*-phrases have no ordinary semantic value but take the set of individuals in their domain as their alternative denotations, as independently proposed earlier by Ramchand (1997):

(14) a. $[\![\text{who}]\!]^{\text{o}}$ undefined b. $[\![\text{who}]\!]^{\text{alt}} = \{x : x \text{ animate}\} = \{\text{ZS, LS, WW, ...}\}$

A clause including 'who' will end up with a denotation akin to our (11) above: no ordinary semantic value, but a non-singleton set of propositions as its alternative set. The application of Q to this structure yields an interpretable *wh*-question. Beck and Kim 2006 extends this approach to the interpretation of alternative questions, with disjunctions projecting alternatives that are interpreted by Q, which is a precursor to my analysis of *háishi* alternative questions. In the discussion that follows, we will see further parallels between *háishi* disjunctions and *wh*-phrases in Mandarin that support this parallel analysis.

Finally, I note that I will ultimately argue that some but not all speakers treat *háishi* as syntactically requiring association with Q. Evidence for this position will come from patterns of non-interrogative uses of *háishi* in section 4.

2.2 Logical disjunctions with *huòzhe*

Next, I discuss the application of existential closure over the alternatives introduced by J, which I propose the standard disjunctor *huòzhe* to require. Here I introduce two variants of an abstract, unary existential closure operator $\exists_{\text{reset}}$ and $\exists_{\text{pass}}$ in (15–16) below. The two operators differ in the resulting alternative set denotation that they provide: $\exists_{\text{reset}}$ returns the singleton set of its ordinary value as the new alternative set, in (15b) — an effect which Beck (2006) describes as "resetting" — whereas $\exists_{\text{pass}}$ simply passes up its sister's alternative set, in (16b).

(15) **The resetting existential operator:**

a. $[\![\exists_{\text{reset}}\ \alpha]\!]^{\text{o}} = \bigvee [\![\alpha]\!]^{\text{alt}}$

b. $[\![\exists_{\text{reset}}\ \alpha]\!]^{\text{alt}} = \left\{\bigvee [\![\alpha]\!]^{\text{alt}}\right\}$

(16) **The passing existential operator:**

a. $[\![\exists_{\text{pass}}\ \alpha]\!]^{\text{o}} = \bigvee [\![\alpha]\!]^{\text{alt}}$

b. $[\![\exists_{\text{pass}}\ \alpha]\!]^{\text{alt}} = [\![\alpha]\!]^{\text{alt}}$

I propose that *huòzhe* is the realization of a J head — with the semantics in (4) — with a syntactic requirement for a corresponding $\exists_{\text{reset}}$ or $\exists_{\text{pass}}$ that quantifies over its alternatives. I also propose that $\exists_{\text{reset}}$ cannot be freely adjoined in the absence of a trigger such as the *huòzhe* J head that requires it, which will become important in the following section. (In contrast, I will argue that $\exists_{\text{pass}}$ *can* be adjoined without a trigger, as I describe and motivate in section 4.2.) I encode this requirement with the uninterpretable feature [u∃] on the J pronounced *huòzhe*, which must be checked by Agree with $\exists_{\text{reset}}$ or $\exists_{\text{pass}}$.[6]

I furthermore propose that ∃ operators adjoin to nodes of propositional type, as in the framework of Kratzer and Shimoyama 2002.[7] We first consider $\exists_{\text{reset}}$, which yields the correct result for examples with a simple disjunction interpretation, such as example (8) with *huòzhe*. The adjunction of $\exists_{\text{reset}}$ at the VP level, based on (9) above, results in the two-dimensional denotation in (17). Notice that this result satisfies

[6] This follows the syntactic treatment of various types of specialized indefinites (e.g. polarity items, free choice items, modal indefinites, etc.) in Kratzer and Shimoyama 2002, Kratzer 2005, Chierchia 2013 (see for example discussion on page 168), and subsequent work. Tsai (2015a) also suggests a similar syntactic feature for *huòzhe* (p. 62). See also Meertens 2021 ch. 8. Note that Agree itself can be long-distance, without syntactic locality restrictions (see e.g. Bošković, 2007; Keine, 2020).

the requirement of Interpretability (3), that $[\![\exists_{\text{reset}}\ \text{VP}]\!]^{\text{o}}$ is defined and is a member of $[\![\exists_{\text{reset}}\ \text{VP}]\!]^{\text{alt}}$.

(17) a. $[\![\exists_{\text{reset}}\ \text{VP}]\!]^{\text{o}} = \textbf{like}(\text{Zhang San}, \text{Li Si}) \vee \textbf{like}(\text{Zhang San}, \text{Wang Wu})$

b. $[\![\exists_{\text{reset}}\ \text{VP}]\!]^{\text{alt}} = \{\textbf{like}(\text{Zhang San}, \text{Li Si}) \vee \textbf{like}(\text{Zhang San}, \text{Wang Wu})\}$

Recall as well that the operator Q which forms alternative questions cannot apply to a structure such as (17), as it already has a defined ordinary value. The result of *huòzhe* with associated $\exists_{\text{reset}}$, as in (17), will therefore necessarily be a logical disjunction, not an alternative question.[8]

Next, we consider the use of $\exists_{\text{pass}}$ to satisfy the [u∃] feature of *huòzhe*. The application of $\exists_{\text{pass}}$ results in the denotation in (18). As noted above, I propose that both $\exists_{\text{reset}}$ and $\exists_{\text{pass}}$ satisfy the syntactic [u∃] requirement of *huòzhe*. However, we notice that (18) does *not* satisfy Interpretability, and therefore by itself is not a grammatical structure. I will argue that the availability of $\exists_{\text{pass}}$ will be useful for explaining the behavior of environments that license non-interrogative uses of *wh*-phrases and (for some speakers) *háishi* disjunction, in section 4.2 below, but concentrate on $\exists_{\text{reset}}$ for the remainder of this section.[9] I note as well that Q cannot apply to form an alternative question from the result of $\exists_{\text{pass}}$ as in (18), as the ordinary value is defined, just as Q cannot apply to the result of $\exists_{\text{reset}}$ as in (17) above.

(18) a. $[\![\exists_{\text{pass}}\ \text{VP}]\!]^{\text{o}} = \textbf{like}(\text{Zhang San}, \text{Li Si}) \vee \textbf{like}(\text{Zhang San}, \text{Wang Wu})$

b. $[\![\exists_{\text{pass}}\ \text{VP}]\!]^{\text{alt}} = \{\textbf{like}(\text{Zhang San}, \text{Li Si}), \textbf{like}(\text{Zhang San}, \text{Wang Wu})\}$

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7 It is also theoretically possible to define existential operators such as $\exists_{\text{reset}}$ and $\exists_{\text{pass}}$ to apply to constituents of non-propositional but conjoinable types, using 'join' (see e.g. Partee and Rooth, 1983). The resulting phrases would then have to take scope via a mechanism such as Quantifier Raising (QR). The proposal for *huòzhe* standard disjunction in Erlewine 2014 also has this effect; see p. 223 note 5 there. I argue later in this section as well as in section 3 that the scope of *huòzhe* disjunction does not pattern as though it is the result of scope-taking movement, contra such QR-based approaches.

8 *Huòzhe* disjunction may however appear in polar questions, whose question radical is a disjunctive proposition; see (i). Concretely, we could assume the polar question particle (PolQ) *ma* has a semantics as in (ii), which requires its sister to have a defined ordinary value. This necessitates the use of $\exists_{\text{reset}}$ as with *huòzhe* before PolQ applies.

(i) Zhāng Sān xǐhuān Lǐ Sì *huòzhe* Wáng Wǔ ma? (based on Dong, 2009: 74)
Zhang San like Li Si SDISJ Wang Wu PolQ
'Does Zhang San like either of Li Si or Wang Wu?'

(ii) a. $[\![\text{PolQ}]\!]^{\text{o}} = \lambda p_{\langle s,t\rangle}\ .\ \{p, \neg p\}$ $\qquad [\![\text{PolQ}]\!]^{\text{alt}} = \{[\![\text{PolQ}]\!]^{\text{o}}\}$

The denotation in (ii) produces a bipolar question denotation, $\{p, \neg p\}$, following Hamblin 1973: 50. I note that Krifka (2015) proposes that "*p ma*" denotes the monopolar $\{p\}$ but see Yuan and Hara 2019 for discussion and defense of the bipolar approach.

9 Both forms of disjunctive denotations — reflecting the application of $\exists_{\text{reset}}$ and $\exists_{\text{pass}}$ in the terms here — are attested in prior work on disjunction in a two-dimensional Alternative Semantics, although this difference has not been explicitly discussed before. For example, the effect of disjunction as proposed in Uegaki 2018 (p. 20) is "resetting" in the sense here, whereas disjunction phrases in von Stechow 1991: 53ff, Romero and Han 2003, and Beck and Kim 2006 project their individual disjuncts as their alternative set denotations. $\exists_{\text{pass}}$ is also adopted in Lohiniva 2019 and Erlewine 2020 for their proposals for alternative unconditionals in Finnish and *wh*-based free choice items in Tibetan, respectively.

I return now to the basic case with $\exists_{\text{reset}}$ in (17) above. As noted above, the subject (here, Zhang San) moves from its predicate-internal position to the surface subject position in Spec,TP. See (19). There are therefore multiple positions of propositional type that $\exists_{\text{reset}}$ could adjoin to. Although this choice of adjunction position does not make an interpretational difference in (19), it can potentially lead to scope ambiguities in more complex sentences.

(19) [$_{\text{TP}}$ Zhang San [λx ... [$_{\text{VP}}$ x like [$_{\text{JP}}$ Li Si *huòzhe*$_{\text{J}}$ Wang Wu]]]]

Consider the examples in (20) below with a quantificational subject or negation. Both sentences are scopally ambiguous in terms of whether the standard disjunction in object position takes scope above or below the higher material. I propose that such contrasts reflect different adjunction positions for $\exists_{\text{reset}}$, for instance to TP or VP as in (19) above.

(20) **Object *huòzhe* disjunction leading to scope ambiguities:**

a. Zài jiǔhuì, { měi-gè rén / hěn-duō rén } chī-le shòusī *huòzhe* yìdàlìmiànshí.
at party every-CL person very-many person eat-PFV sushi SDISJ pasta
'At the party, {everyone / many people} ate sushi or pasta.'
(everyone/many > or, or > everyone/many) (based on Crain, 2012: 258)

b. Zhè-lù chē bù tíng fǎyuàn *huòzhe* túshūguǎn.
this-route bus NEG stop courthouse SDISJ library
'This bus doesn't stop at the courthouse or the library.'
(not > or, or > not) (Jing, 2008: 169–170)

For sentences such as (20b) with standard disjunction in object position with clausemate negation, many prior works report that the wide scope disjunction reading (or > not) is dominant for adult Mandarin speakers.[10] However, Jing (2008) argues that both scopes are possible in sentences such as (20), especially when supported by explicit disambiguating continuations as in (21) below. See also Liu and Chen 2017 for an experimental study showing that adults interpret structures with object *huòzhe* disjunction and clausemate *méi* negation as scopally ambiguous as well.

[10] This is in contrast to the narrow scope disjunction reading which is dominant for the corresponding English structure as well as for Mandarin-acquiring children. See Jing et al. 2005; Crain 2012; Crain and Thornton 2012; Notley et al. 2012; Gao et al. 2018.

(21) **Disambiguating continuations for (20b):** (Jing, 2008: 169–170)

a. Continuation supporting “not > or” parse for (20b):

Rúguǒ	nǐ	yào	qù	zhè	liǎng-gè	dìfāng,	yào	huán	yī-lù	chē.
if	2sg	need	go	this	two-CL	place	need	change	one-route	bus

‘If you need to go to (either of) these two places, you need to change to another bus.’

b. Continuation supporting “or > not” parse for (20b):

Dàn	wǒ	bù	jìdé	tā	bù	tíng	nǎ	zhàn	le.
but	1sg	NEG	remember	3sg	NEG	stop	which	stop	LE

‘But I don’t remember which one it doesn’t stop at.’

It is important to note that the scope ambiguities observed in examples such as (20a,b) argue against an approach where the standard disjunction *huòzhe* forms quantificational phrases that then take scope as other object quantifiers do, for instance via Quantifier Raising (QR) or a similar mechanism. Mandarin Chinese is well known as a scope-rigid language (see e.g. Huang, 1982; Aoun and Li, 1989). In particular, experimental studies confirm that universal quantifiers in object position cannot scope over quantificational subjects (Scontras et al., 2017) nor over negation (Fan, 2017) for most Mandarin-speaking adults. The scope-taking behavior of standard disjunction as in (20a,b) is thus better modeled as reflecting flexibility in the adjunction positions of $\exists_{\text{reset}}$, as I propose here. I also present additional evidence on the scope-taking behavior of standard disjunction *huòzhe* which will further motivate this approach in section 3 below.

2.3 Summary

I propose that the standard disjunctor *huòzhe* and the interrogative disjunctor *háishi* share a basic syntax as a junction head J, which can take disjuncts of different categories and sizes. Adopting the two-dimensional framework of Rooth-Hamblin Alternative Semantics, J has the semantics of a set-union operator in its alternative set dimension, resulting in an alternative set denotation for JP as the set of its disjuncts’ denotations. The JP then composes pointwise with additional syntactic material above it. This results in JP-containing VP and TP meanings that have an alternative set denotation as a set of propositions, but no defined ordinary value, which then cannot be interpreted without a higher interpreting operator.

I presented two different ways that JP-containing clauses can be productively interpreted. The first is with the operator Q (as in Beck, 2006; Beck and Kim, 2006; Kotek, 2019), resulting in an alternative question denotation. The second is with the existential operator $\exists_{\text{reset}}$ which results in an interpretable disjunctive proposition. (I also introduced a variant, $\exists_{\text{pass}}$, which I motivate and further discuss in section 4.2 below.) I propose that the standard disjunctor *huòzhe* bears a syntactic feature [u$\exists$] requiring an associated $\exists$ operator, which in turn blocks the application of Q to evaluate its alternatives to form a question. In turn, I propose that the interrogative disjunctor *háishi* lacks this feature and that adjunction of $\exists_{\text{reset}}$ must be motivated, thereby leaving interpretation with Q as an alternative question as the only viable path for its interpretation, in simple cases. This derives the observed one-to-one correlation between disjunction form and clause type in simple cases. In addition, we might consider describing *háishi* as having a syntactic feature such as Huang et al.'s 242 (2009) [+wh] to further distinguish it from *huòzhe*; I will argue in section 4 below that some but not all Mandarin speakers have an extra syntactic feature on *háishi* to restrict its distribution.

3 Motivating evidence

I now elaborate on various details regarding the behavior and interpretation of the two disjunctors, which serve to motivate my particular proposal above in relation to possible alternative accounts, including those in prior literature. I will show that both forms of disjunction are insensitive to syntactic islands (§3.1) but are subject to so-called focus intervention effects (§3.2). Island-insensitivity serves to argue against analyses of both forms of disjunction that involve covert movement for their scope-taking, as J. Huang (1982) originally proposed for *háishi*. I will emphasize as well that the two forms of disjunction neatly parallel one another in their insensitivity to syntactic islands and sensitivity to focus intervention, supporting my approach where both disjunctors project alternatives, in the same manner, up to their interpreting operators.

3.1 Island insensitivity

The study of the syntax and semantics of Mandarin Chinese has played a starring role in early discussions of covert movement. As Mandarin Chinese is a *wh*-in-situ language, James Huang (1982) argued that some *wh*-words move covertly to the corresponding interrogative complementizer at Logical Form (LF) without affecting the word order. These movements, although not reflected in word order, are nonetheless detected by their sensitivity to island constraints on movement (Ross, 1967).

James Huang also suggested in passing in this early work that *háishi* alternative questions may involve covert movement of *háishi* to the interpreting complementizer (Huang, 1982: 276);[11] concretely,

[11] This suggestion appears to be intended as just one possible analysis, however, with another being that *háishi* coordinates disjuncts

we might imagine this to suggest that the entire *háishi*-headed JP moves covertly. However, he later argued against the idea of covert movement for *háishi* after demonstrating that *háishi* disjunction is *not* sensitive to syntactic islands (J. Huang, 1991). That is, in the terms of my account here, a JP headed by the interrogative disjunctor can be embedded inside an island and still lead to the interpretation of a surrounding clause as an alternative question. Here I reproduce a few of James Huang's examples involving sentential subject islands and relative clause islands in (22–23) below, supplemented with examples from Ray Huang's work demonstrating insensitivity to complex NP islands and adjunct islands in (24–25). All of these examples are alternative questions. The relevant island structures and *háishi* JPs are indicated below.

(22) **Interrogative disjunction is not sensitive to sentential subject islands:** (J. Huang, 1991: 313)

a. $[_{\text{island}}$ Wǒ $[_{\text{JP}}$ [qù měiguó] *háishi* [bú qù měiguó]]] bǐjiào hǎo?
1sg go America IDISJ NEG go America comparatively good
'Is it better for me to go to America or to not go to America?'

b. $[_{\text{island}}$ $[_{\text{JP}}$ Wǒ *háishi* nǐ] qù měiguó] bǐjiào hǎo?
1sg IDISJ 2sg go America comparatively good
'Is it better for me to go to America or for you to go to America?'

(23) **Interrogative disjunction is not sensitive to relative clause islands:**

a. Nǐ xǐhuān [$[_{\text{island}}$ ___ $[_{\text{JP}}$ [zūnzhòng nǐ] *háishi* [bù zūnzhòng nǐ]] de] rén]?
2sg like respect 2sg IDISJ NEG respect 2sg DE person
'Do you like people who respect you or people who don't respect you?' (J. Huang, 1988a: 688)

b. Nǐ xǐhuān [$[_{\text{island}}$ $[_{\text{JP}}$ Zhāng Sān *háishi* Lǐ Sì] xiě ___ de] shū]?
2sg like Zhang San IDISJ Li Si write DE book
'Do you like the books that Zhang San wrote or the books that Li Si wrote?' (R. Huang, 2010a: 123)

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of clausal size which undergo Conjunction Reduction; see his example (221) and preceding prose on page 276. I discuss and argue against this alternative approach in Appendix A.

(24) **Interrogative disjunction is not sensitive to complex NP islands:** (R. Huang, 2010a: 125–126)

a. Nǐ xiāngxìn [[$_{\text{island}}$ Xiǎodí shì [$_{\text{JP}}$ [yīnwèi qiàn zhài] *háishi* [yīnwèi shī liàn]]
2sg believe Xiaodi SHI because owe debt IDISJ because lose romance

ér zìshā de] shuōfǎ] ne?
so suicide DE story NE

'Do you believe the story that Xiaodi committed suicide because of owing debt or because of failing at love?'

b. Nǐ xiāngxìn [[$_{\text{island}}$ [$_{\text{JP}}$ [tā dé jiǎng] *háishi* [wǒ dé jiǎng] de] xiāoxi] ne?
2sg believe 3sg get prize IDISJ 1sg get prize DE news NE

'Do you believe the news that he/she won the prize or that I won the prize?'

(25) **Interrogative disjunction is not sensitive to adjunct island:** (R. Huang, 2020: 211)

Nǐ [$_{\text{island}}$ zài Zhāng Sān [$_{\text{JP}}$ [kāi-le dēng] *háishi* [guān-le dēng]] zhīhòu] cái
2sg at Zhang San turn.on-PFV light IDISJ turn.off-PFV light after then

jìn fángjiān?
enter room

'Did you enter the room after Zhang San turned on the light or after Zhang San turned off the light?'

Wh-in-situ in Mandarin Chinese exhibits an argument/adjunct asymmetry, whereby only *wh*-adjuncts exhibit sensitivity to island effects (Huang, 1982). But we note that the island insensitivity of *háishi* interrogative disjunction is not contingent on the type of material that the JP represents. We see that interrogative disjunctions of arguments (as in (22b) and (23b)), adjuncts (as in (24a)), as well as of clausal/verbal extended projections (as in (22a), (23a), (24b), and (25)) are all insensitive to syntactic islands.

This is not to say that the embedding of interrogative disjunction is entirely unconstrained. R. Huang (2010a: 138, 2010b: 227) and He (2011) show that interrogative disjunction is ungrammatical inside an appositive relative clause:[12]

[12] The relative clauses here in precedes a deictic demonstrative and hence must be appositive; see Sun and Lai 2019 and discussion there. Example (i) below, also from He 2011, shows that it is possible to form an alternative question with two disjuncts that appear on the surface to differ only in the content of their appositive relative clauses, although in this case the intended reading involves a choice between two different books with separate deictic indices.

(26) **Interrogative disjunction ungrammatical in appositive relative clause:** (He, 2011: 90)

*Nǐ zuì xǐhuān [$_{\text{appositive}}$ [$_{\text{JP}}$ Zhāng Sān *háishi* Lǐ Sì] xiě ___ de] nà-běn shū?
2sg most like Zhang San IDISJ Li Si write DE that-CL book
literally: 'Do you like that book, which Zhang San or Li Si wrote, the most?'

Del Gobbo (2010: 403–405, 2015: 76–78) has shown that various other question-introducing constructions are ungrammatical in Mandarin appositive relatives. This restriction appears to reflect a more general constraint that appositive content is not at-issue (see e.g. AnderBois et al., 2015), i.e. does not address or raise new questions under discussion, although certain exceptions are reported in some other languages; I refer the interested reader to discussions in these works and the citations there.

In summary, the interpretation of *háishi* interrogative disjunction leading to an alternative question is insensitive to classic syntactic islands, which are diagnostic of syntactic movement, but they *do* exhibit restrictions based on their semantics. This supports my proposal whereby *háishi* interrogative disjunctions do not undergo movement in alternative questions, contrary to the early suggestion by J. Huang (1982: 276).[13]

I now turn to the interpretation of *huòzhe* standard disjunctions, which under my account should echo *háishi* interrogative disjunctions in their interpretational possibilities under embedding. Under my account, the core difference between declaratives with *huòzhe* and structurally parallel alternative questions with *háishi* is that the former involves a covert operator $\exists_{\text{reset}}$ to interpret the disjunction's alternatives, whereas the latter has Q as the corresponding operator. The structural position of $\exists_{\text{reset}}$ determines the scope that the *huòzhe* standard disjunction can be described to take.

In contrast to the above discussion of *háishi* alternative questions, to my knowledge no prior work has systematically investigated the interaction of *huòzhe* standard disjunction and syntactic islands. Consider example (27) below. A *huòzhe* standard disjunction is embedded within a relative clause island. This target sentence is followed by a disambiguating continuation of the form discussed in Jing 2008 (see (21b) above), which is incompatible with a parse of the preceding sentence where disjunction takes scope within

(i) [$_{\text{JP}}$ [Nǐ zuì xǐhuān [$_{\text{appositive}}$ Zhāng Sān xiě ___ de] nà-běn shū] *háishi* (He, 2011: 90)
2sg most like Zhang San write DE that-CL book IDISJ

[nǐ zuì xǐhuān [$_{\text{appositive}}$ Lǐ Sì xiě ___ de] nà-běn shū]]?
2sg most like Li Si write DE that-CL book

'Do you like that book, which Zhang San wrote, the most or do you like that (other) book, which Li Si wrote, the most?'

[13] More recently, R. Huang (2020) has proposed that Mandarin alternative questions *do* involve covert movement, but with the possibility of large covert pied-piping, thereby obscuring any island sensitivity. The description there makes the proposed covert movement difficult to detect or falsify. R. Huang (2020) furthermore does not provide a compositional semantics, alluding only to syntactic feature percolation between the disjunctor and the covertly moved phrase.

the island.[14]

(27) **Standard disjunction is not sensitive to relative clause islands:**

Wǒ	zhīdào	tā	mǎi-le	[yī-běn	[$_{\text{island}}$ [$_{\text{JP}}$	Zhāng Sān	*huòzhe*	Lǐ Sì]	xiě	___]	de	shū],
1sg	know	3sg	buy-PFV	one-CL		Zhang San	SDISJ	Li Si	write		DE	book

dànshì	wǒ	bù	zhīdào	shì	Zhāng Sān	háishi	Lǐ Sì	xiě	de.
but	1sg	NEG	know	COP	Zhang San	IDISJ	Li Si	write	DE

‘I know he/she bought a book that Zhang San or Li Si wrote, but I don’t know if it was the one that Zhang San or Li Si wrote.’

On my account here, this scope-taking reflects the adjunction position of $\exists_{\text{reset}}$ in the matrix clauses, with alternatives introduced by the *huòzhe* disjunction JPs composing pointwise with the material in the island and above, up to $\exists_{\text{reset}}$. This propagation of alternatives is not sensitive to syntactic islands. This example also serves to show that the relationship between *huòzhe* and its corresponding $\exists$ operator is not restricted by intervening clause boundaries; see note 6 above on a related syntactic detail.

Finally, I note that Erlewine (2014: 226) and R. Huang (2020: 234) report that *háishi* interrogative disjunction is sensitive to *wh*-islands; that is, *háishi* inside an embedded question cannot scope out to itself raise a matrix alternative question. I discuss such effects in Appendix B, where I show that the facts are more complicated than what has been described in these prior works. Importantly, however, the restrictions on alternative question formation with *háishi* again parallel restrictions on the scope-taking with *huòzhe* standard disjunction in the same environments, supporting my overall account.

In summary, I have shown that the scope-taking behavior of the standard disjunction is thus insensitive to syntactic islands, just as we saw with *háishi* interrogative disjunction above. Neither the behavior of alternative questions with *háishi* nor of *huòzhe* standard disjunction suggests that their derivations involve covert movement for their scope-taking.

3.2 Focus intervention effects

Although the projection of alternatives in Rooth-Hamblin Alternative Semantics is insensitive to syntactic barriers, it has been argued to be susceptible to so-called *focus intervention effects* (Beck, 2006; Beck and Kim, 2006, and others). In this section, building on observations reported in Erlewine 2014 and Li and Law 2016, I will show that both *háishi* interrogative disjunction in alternative questions as well as

[14] Under certain circumstances, quantifiers in Mandarin may take scope out of relative clauses via a mechanism such as QR. However, this is not the case when the relative clause follows the numeral classifier, as in (27) here. See discussion in Wang 2023: 41–42, 2025: ch. 2.

the scope-taking possibilities of *huòzhe* standard disjunction are susceptible to focus intervention effects. This again supports my proposal where both forms of disjunction share a common function of projecting alternatives from their surface position, which are then interpreted by corresponding alternative-sensitive operators Q and $\exists_{\text{reset}}$ in the basic cases.

So-called focus intervention effects most commonly refer to the relative ungrammaticality of *wh*-questions where an "intervener" intervenes between an in-situ *wh*-word and its interpreting operator, Q. Consider the contrast in (28) below. Even though Mandarin is a *wh*-in-situ language, the object *wh*-question in (28a) is judged as degraded due to the subject 'only.' However, if the *wh*-phrase is fronted across the subject as in (28b) or is pseudoclefted as in (28c), the question is grammatical with the intended interpretation.[15] This reflects the status of the pre-subject 'only' (in bold) as a problematic intervener. See also Yang 2008, 2012, Li and Cheung 2015, Li and Law 2016, among others, for further data and discussion of focus intervention effects in Mandarin Chinese *wh*-questions.

(28) **Focus intervention effect with subject 'only':** (a,b based on Kim, 2006: 166)

a. ?* **Zhǐyǒu** [Zhāng Sān]$_F$ kàn-le *nǎ-běn* *shū* (ne)?
only Zhang San read-PFV which-CL book NE

b. (Shì) *Nǎ-běn* *shū*, **zhǐyǒu** [Zhāng Sān]$_F$ kàn-le *t* (ne)?
SHI which-CL book only Zhang San read-PFV NE
'Which book did only Zhang San read?'

c. [**Zhǐyǒu** [Zhāng Sān]$_F$ kàn *t* de] shì *nǎ-běn* *shū* (ne)?
only Zhang San read DE COP which-CL book NE
'Which book is the one that only Zhang San read?'

In one prominent proposal, Kim (2002, 2006) and Beck (2006) argue that the problematic interveners in many languages share the property of being focus-sensitive operators. In the standard Roothian Alternative Semantics for focus particles, focus-sensitive operators such as 'only' have the effect of "resetting" their alternative sets, in the sense introduced in section 2.2 above, blocking the projection of the *wh*-phrase's alternatives up to the intended interpreting operator, Q, at the edge of the question. I comment further on one complication regarding focus intervention according to Beck's account at the end of this section, but for the discussion that follows, I will simply use focus intervention effects — especially with intervening 'only' as in (28) — as a diagnostic for the projection of Rooth-Hamblin alternatives.

[15] Cheung (2014) argues that the *shì* that often precedes fronted *wh*-phrases is the focus marker *shì* (see e.g. Huang, 1982, 1988b, as well as Erlewine 2022), rather than the copula. I therefore gloss *shì* in (28b) and (30b) as SHI but gloss *shì* in pseudoclefts such as (28c) and (30c) as a copula, COP.

Beck and Kim (2006) show that such focus intervention effects also affect alternative questions in many languages. Example (29a) is ungrammatical as an alternative question in English, although the same alternative question without 'only' is grammatical. According to Beck and Kim, the disjunction projects alternatives, which must be interpreted by Q at the clause edge for its grammatical alternative question interpretation, but the presence of 'only' in (29a) blocks this projection. Similar to the contrast in (28) above, fronting the alternative source (here, the disjunction) out of the scope of the intervener as in (29b) results in a grammatical alternative question with the same intended interpretation.

(29) **Focus intervention in English alternative questions:** (Beck and Kim, 2006: 167)

a. *Does **only** [John]$_F$ like [Mary *or* Susan]?

b. Is it [Mary *or* Susan] that **only** [John]$_F$ likes *t*?

Returning now to our Mandarin disjunctors, Erlewine (2014) shows that interrogative disjunction is subject to focus intervention effects, as in (30). As with the Mandarin *wh*-question (28) or the English alternative question (29), fronting the disjunction out of the scope of 'only' (30b) (with preceding focus marker *shì*; see note 15) or pseudoclefting it (30c) results in grammatical alternative questions with the same intended interpretation. This suggests that the ungrammaticality of (30a) reflects a focus intervention effect of the Beck and Kim type, caused by a focus-sensitive operator hierarchically coming between the interrogative disjunction JP and its corresponding Q at the clause edge. Intervention in Mandarin alternative questions is also reported with the focus marker *shì* (He, 2011: 87; Erlewine, 2014: 228) and with the high negator *búshì* (R. Huang, 2020: 214), but here I concentrate on intervention with 'only.'

(30) **Focus intervention in Mandarin alternative questions:**

a. ***Zhǐyǒu** [Zhāng Sān]$_F$ chī-le [$_{JP}$ píngguǒ *háishi* júzi] (ne)?
only Zhang San eat-PFV apple IDISJ orange NE
Intended: 'Was it an apple or an orange that only Zhang San ate?' (Erlewine, 2014: 228)

b. Shì [$_{JP}$ píngguǒ *háishi* júzi], **zhǐyǒu** [Zhāng Sān]$_F$ chī-le *t* (ne)?
SHI apple IDISJ orange only Zhang San eat-PFV NE
'Was it an apple or an orange that only Zhang San ate?'

c. [**Zhǐyǒu** [Zhāng Sān]$_F$ chī *t* de] shì [$_{JP}$ píngguǒ *háishi* júzi] (ne)?
only Zhang San eat-PFV DE COP apple IDISJ orange NE
'Was what only Zhang San ate an apple or an orange?'

Another way that this intervention effect can be avoided is by using larger disjuncts which each individually contain the focus-sensitive operator, as in (31) below. In this case, although the contrasting constituents 'apple' and 'orange' are each in the scope of the intervener 'only,' there is no intervener between the JP and the edge of the clause. It is the region between the JP and the interpreting operator Q which is susceptible to intervention, explaining the difference between the local object disjunction in (30a) and the clausal disjunction in (31).

(31) **'Only' in each disjunct does not trigger intervention; cf (30a):**

[$_{\text{JP}}$	[Zhǐyǒu	[Zhāng Sān]$_{\text{F}}$	chī-le	píngguǒ	(ne)]	*háishi*	[zhǐyǒu	[Zhāng Sān]$_{\text{F}}$	chī-le	júzi	(ne)]]?
	only	Zhang San	eat-PFV	apple	NE	IDISJ	only	Zhang San	eat-PFV	orange	NE

'Did only Zhang San eat-PFV an apple or did only Zhang San eat an orange?'

Next we turn to the scope-taking of *huòzhe* standard disjunction. Under my account, the scope-taking possibilities of standard disjunction in Mandarin should also be constrained by focus intervention effects: JPs headed by *huòzhe* project alternatives, just as those headed by *háishi* do, and must be interpreted by an existential operator (in the basic case, $\exists_{\text{reset}}$) which determines the scope of the disjunction.[16]

Crain (2012) reports that *huòzhe* standard disjunction cannot scope over subject 'only' as in example (32) below. We can verify that the wide scope disjunction reading is not available with the context described below, where the sentence should be judged to be felicitous and true if the wide scope disjunction reading — that only John ate an apple or only John ate a pear — were available, because in this situation, it is true that only John ate a pear. However, the sentence is judged as false for all but one of the speakers that I have consulted. Xie (2020: 17) also reports this same scope restriction in parallel sentences as well.

(32) **Scope of standard disjunction restricted by subject 'only':** (Crain, 2012: 242–243)

Zhǐyǒu	[Yuēhàn]$_{\text{F}}$	chī-le	[$_{\text{JP}}$ píngguǒ	*huòzhe*	lí	].
only	John	eat-PFV	apple	SDISJ	pear	

'Only John ate an apple or a pear.' (only > or, *or > only)

False in context where John ate an apple and a pear and Mary ate an apple and an orange.

Recall that *huòzhe* standard disjunction in object position can generally lead to scope ambiguities, including with respect to quantificational subjects, as we saw with example (20a) above. As discussed there, such scope ambiguities reflect different adjunction positions for $\exists_{\text{reset}}$, for instance to TP or VP, as schematized in (33) below:

[16] Li and Law (2016) also present evidence of focus intervention effects affecting what I will analyze below as interpretation by $\exists_{\text{pass}}$. See discussion in section 4.2 below.

(33) ($^{\checkmark}\exists_{\text{reset}}$) [$_{\text{TP}}$ everyone [λx ... [($^{\checkmark}\exists_{\text{reset}}$) [$_{\text{VP}}$ x ate [$_{\text{JP}}$ sushi *huòzhe*$_{\text{J}}$ pasta]]]]] for (20a)

High $\exists_{\text{reset}} \Rightarrow$ or > everyone; low $\exists_{\text{reset}} \Rightarrow$ everyone > or

Given the general availability of $\exists_{\text{reset}}$ in these two positions as in (33), the unavailability of the wide scope disjunction reading in (32) deserves an explanation. We can understand this too as a focus intervention effect, following discussion in Li and Law 2016: 225–226. The JP projects alternatives which will compose pointwise with higher material and must be interpreted by the corresponding $\exists_{\text{reset}}$ for the intended interpretation. If the focus-sensitive operator 'only' intervenes, this causes an intervention effect of the Beck and Kim sort, and therefore only the low adjunction of $\exists_{\text{reset}}$ is available here.

(34) ($^{*}\exists_{\text{reset}}$) [$_{\text{TP}}$ only [John]$_{\text{F}}$ [λx ... [($^{\checkmark}\exists_{\text{reset}}$) [$_{\text{VP}}$ x ate [$_{\text{JP}}$ apple *huòzhe*$_{\text{J}}$ orange]]]]] for (32)

Finally, I note that not all instances of disjunction within the scope of a focus-sensitive operator are ungrammatical. Specifically, it is a focus particle associating with a separate focus — in these examples here, the subject — which causes intervention, but a focus particle associating *with* a *wh*-phrase or an interrogative disjunction does not block their intended question interpretations, as in (35a,b) below. Similarly, focus association with the *huòzhe* standard disjunction in (35c) below allows for the disjunction to scope over the 'only.' Li and Law (2016) claim that such facts challenge the Beck 2006 theory of focus intervention, as well as the overall framework of Rooth-Hamblin Alternative Semantics, which treats the alternatives from focus and the alternatives of *wh*-phrases and disjunction as the same ontological objects.

(35) **Focus association *with* the *wh* or disjunction does not trigger intervention:**

a. Tā **zhǐ** xǐhuān *shéi*?
3sg only like who
≈ 'Who x is such that he/she only likes x?' (Aoun and Li, 1993: 207)

b. Lǐ Bái **zhǐ** hē-le [$_{\text{JP}}$ kāfēi *háishi* hóngchǎ]?
Li Bai only drink-PFV coffee IDISJ tea
≈ 'Is it x tea or coffee such that Li Bai drank only x?' (Li and Law, 2016: 230)

c. Yuēhàn **zhǐ** chī-le [$_{\text{JP}}$ píngguǒ *huòzhe* lí].
John only eat-PFV apple SDISJ pear
'John only ate an apple or a pear.' (only > or)
'John only ate an apple or John only ate a pear.' (or > only) (based on Li and Law, 2016: 227)

In recent work (Erlewine, to appear), I have argued that such facts do *not* threaten the Rooth-Hamblin framework, contrary to the conclusion of Li and Law 2016. There, I propose an independently motivated and modest revision to the Rooth 1992 approach to focus association, which allows us to correctly interpret examples of this form while also maintaining the ungrammaticality of focus intervention configurations as in (28) and (30) above. For reasons of space, I will not detail this proposal here, but I refer the reader to (30–32) in Erlewine to appear for the interpretation of an example parallel to (35a) above, the logic of which also extends to the interpretation of (35b,c). This approach also extends to cases where the focus marker *shì* associates with the alternative source in constituent questions (Huang 1982; Erlewine 2014; Xu 2010; see also note 15), as is especially common with *háishi* alternative questions. Note as well that Li and Law (2016) must additionally assume that disjunctions but not *wh*-phrases bear F-marking in these configurations (see discussion in their pp. 233–234, as well as their note 16), whereas the *wh* and *háishi* variants of these configurations can be treated more uniformly under the approach in Erlewine to appear. I refer interested readers to Erlewine to appear for full discussion.

3.3 Summary

In this section, I discussed potential restrictions on the interpretation of the two disjunctors. We have seen that alternative question formation with *háishi* interrogative disjunction as well as the scope-taking of *huòzhe* standard disjunction are insensitive to syntactic islands (§3.1), which affect overt and covert movement (J. Huang, 1982), but are sensitive to focus intervention effects (§3.2). The island-insensitivity facts specifically argue against prior proposals that involve the covert movement of *háishi* in alternative questions (as suggested in J. Huang 1982: 276) or the covert movement of *huòzhe* disjunctions for their scope-taking (as suggested in Erlewine 2014: 223 note 5). The observed restrictions on their interpretation are instead what is predicted by my account, where both types of disjunctions project Roothian alternatives in a manner that leaves them susceptible to so-called focus intervention effects as in Beck 2006 and Beck and Kim 2006.

Throughout this section, we have seen that the two disjunctors precisely parallel one another. This in turn supports my proposal here, that both disjunctors share a common core, J, taking disjuncts of variable syntactic size and projecting their these disjuncts as alternatives.[17] The only difference between them is the choice of corresponding operator that quantifies over their alternatives: in the simple cases discussed so far, Q for *háishi* interrogative disjunction, yielding alternative question meanings, and $\exists_{\text{reset}}$ for *huòzhe* standard disjunction, yielding a boolean disjunction meaning. Having established this deep uniformity

[17] I refer readers to Appendix A for syntactic arguments that *háishi* interrogative disjunction takes disjuncts of variable size, contrary to the proposals in J. Huang et al. 2009 and R. Huang 2009, 2010a,b.

between the two disjunctors, I now turn to environments that complicate this one-to-one correlation, which will further inform our understanding of the nature of this link between disjunction form and their corresponding operators.

4 Non-interrogative uses of *háishi*

In this paper, I have adopted the terms "interrogative disjunction" versus "standard disjunction" from Haspelmath (2007) and Mauri (2008) to describe the difference between the Mandarin disjunctors *háishi* and *huòzhe*. Indeed, in all examples seen thus far (excepting (2) in the introduction), the use of *háishi* leads to a natural alternative question interpretation, whereas *huòzhe* leads to the formation of a disjunctive proposition. However, as previewed in the introduction, there are certain situations where the use of *háishi* does *not* give rise to an alternative question. These non-interrogative uses of *háishi*, and their connection to non-interrogative uses of *wh*-words, will be the subject of this section.

I will organize the discussion here by classifying non-interrogative uses of *háishi* into three classes: unconditional and related universal uses, as in (36); wide-scope existential uses with speaker ignorance inferences, as in (37); and existential uses in certain licensing environments, such as with negation in (38). Notably, there is substantial variation between speakers in the availability of this third class of uses, as indicated here by the % judgement mark on (38).

(36) **Universal *háishi* via unconditionals:** (Lü, 1980: 173)

Wúlùn shàngbān *háishi* xiūxí, tā dōu zài zhuómó xīn-de shèjì fāng'àn.
no.matter at.work IDISJ rest 3SG DOU PROG polish new-DE design plan
'Both when at work and resting, he/she is always crafting new design plans.'

(37) **Existential *háishi* with speaker ignorance/irrelevance:**
(Yuan, 2021: 73, based on a corpus example)

Máo zhǔxí zài 1943-nián *háishi* 1944-nián gěi tā xiě-guò yī-ge jiǎngzhuàng...
Mao chairman LOC 1943-year IDISJ 1944-year to 3SG write-EXP one-CL certificate

Jùtǐ nǎ-nián jì-bù-qīng-le.
concrete which-year remember-NEG-clear-PFV

'Chairman Mao had written him a certificate of merit in 1943 or 1944... I can't remember exactly which year.'

(38) **Existential *háishi* under negation, for some speakers:** (Lin H.-Y., 2008: 1)

%Tā	méiyǒu	mǎi	píngguǒ	*háishi*	lízi.
3sg	NEG.PFV	buy	apple	IDISJ	pear

'He/she didn't buy apples or pears.'

As I will establish in greater detail below, Mandarin speakers fall broadly into two groups, which I will refer to as Type A and Type B here. Type A speakers are more restrictive, allowing for non-interrogative *háishi* only in the unconditional and related universal uses as in (36) and wide-scope existential uses as in (37). Type B speakers also allow the narrow-scope existential uses under certain licensing operators, as in (38). As Lin Hsin-yin (2008) observes, the environments and interpretations for non-interrogative *háishi* for Type B speakers line up with those of non-interrogative uses of *wh*-words in the language, which includes both universal and existential uses (see e.g. J.-W. Lin, 1996, 1998b).[18]

I will analyze these non-interrogative uses and the attested speaker variation as follows. First, I observe that the universal uses of *háishi* as in (36) are essentially embedded question environments involving Q, building on previous work on the analysis of unconditionals with *dōu* (Cheng and Huang, 1996; Lin, 1996). At the same time, I observe that the wide-scope existential use as in (37) is as an *epistemic disjunction* with speaker ignorance inferences; I follow previous work on epistemic indefinites (see e.g. Alonso-Ovalle and Menéndez-Benito, 2013, 2015) which derive their speaker ignorance inferences from covert operators at the sentence root. I therefore propose that *háishi* for Type A speakers has a syntactic feature [uR] (R for "root") that must be checked by association with a relevant operator in the high, clause-typing domain of the clause (39a), which limits the use of *háishi* to precisely these contexts.

In contrast, I propose that *háishi* does *not* bear [uR] for Type B speakers; see (39b) below. As previewed in section 2 above, the result of *háishi* disjunction is a semantic object akin to that of a *wh*-phrase.

[18] The Type A vs Type B classification appears to roughly — but not entirely — correspond to a dialectal distinction between Mandarin speakers in mainland China versus Taiwan, respectively. This regional association is suggested by Ray Huang, in personal correspondence to Bhadra 2017 (p. 171 note 2), as well as by two of my anonymous reviewers. Indeed, the native speaker linguists who have reported Type B pattern judgments in prior literature are all from Taiwan: Lin Hsin-yin (2008), Hsieh Miao-Ling (2004), Ray Huang (2010a), and Edwin Tsai (2015a). In contrast, Yuan Mengxi (2021) conducts an acceptability judgment experiment with 175 mainland Chinese speakers, as well as a study of mainland Chinese corpora, and reports the pattern of judgments that I call Type A here; I elaborate on her findings in section 4.3. Nevertheless, some Taiwanese speakers that I and a reviewer have consulted also appear to command more restrictive, Type A grammars, and I have also identified Type B speakers not from Taiwan, so the two types cannot be neatly described as a dialectal distinction.

Si-Kai Lee (p.c.) suggests that, if Type B grammars are indeed more likely in certain regions such as Taiwan, a relevant factor may be the fact that Taiwanese Southern Min (TSM) does not share the feature of having distinct interrogative versus standard disjunctors as Mandarin does; that is, TSM utilizes one disjunctor, variably realized as *i̍ah-sī* or *a̍h-sī*, for the expression of alternative questions as well as boolean disjunction (see e.g. Hsiao and Her, 2021: 264–265 note 8). Although this is an intriguing possible connection, it would not immediately explain the behavior of the Type B Mandarin grammar, given that non-interrogative *háishi* is still limited to particular environments; that is, it is not simply the case that the distinction between the two forms is collapsed.

With the absence of a [uR] specification — just like *wh*-phrases, which lack such a syntactic restriction for all speakers — we correctly predict *háishi* for Type B speakers to have non-interrogative uses in the same range of environments that *wh*-phrases do.

(39) **A featural difference between Type A and Type B speakers:**

a. Type A: *háishi* = [J, uR]; *huòzhe* = [J, u∃]
where [uR] can be checked by association with Q or $\exists_R$ (to be introduced in section 4.3)

b. Type B: *háishi* = [J]; *huòzhe* = [J, u∃]

In the following sections, I first discuss unconditionals and related *dōu* constructions in section 4.1, then the narrow-scope existential uses for Type B speakers in section 4.2, and then return to the wide-scope existential uses with speaker ignorance and irrelevance in section 4.3.

4.1 *Háishi* in unconditionals and related *dōu* constructions

I begin by describing the uses of *háishi* disjunctions in so-called unconditionals, available to all speakers (Type A and B). The term "unconditional" refers to constructions that express that a particular consequent will hold across a set of possible circumstances (Zaefferer, 1991). Unconditionals may involve a *wh*-phrase or a disjunction, as exemplified by the English in (40).

(40) **English unconditionals:**

a. No matter which route we take, we'll get to the beach eventually. (Rawlins, 2013: 113)

b. Whether we take route A or route B, we'll get to the beach eventually.

Unconditionals in Mandarin may include a *wh*-phrase or an A-not-A verb form, as in (41). As indicated here, these unconditional adjuncts may be introduced by an expression such as *wúlùn* or *bùguǎn*, which both literally echo the English *no matter*. Although *wh*-phrases in Mandarin have a range of non-interrogative uses (discussed further in the following sections), the A-not-A verb form is strongly associated with the formation of polar questions, leading Cheng and Huang (1996: 147–149) and Lin (1996: 76–77) to describe Mandarin unconditionals as involving question embeddings.

(41) **Mandarin unconditionals:** (based on Lin, 1996: 76–77)

a. [$_{\text{uncond}}$ (Wúlùn/bùguǎn) nǐ yāoqǐng shéi], wǒ dōu huānyíng tā.
no.matter 2sg invite who, 1sg DOU welcome 3sg
'No matter who you invite, I will welcome him/her.'

b. [$_{\text{uncond}}$ (Wúlùn/bùguǎn) nǐ qù-bú-qù], wǒ dōu yào qù.
no.matter 2sg go-NEG-go 1sg DOU want go
'No matter whether you go or not, I want to go.'

Against this backdrop, it is unsurprising that Mandarin unconditionals can involve the interrogative disjunction *háishi*, as in (42) below.

(42) **Mandarin unconditional with *háishi*:** (Chen, 2022: 98)

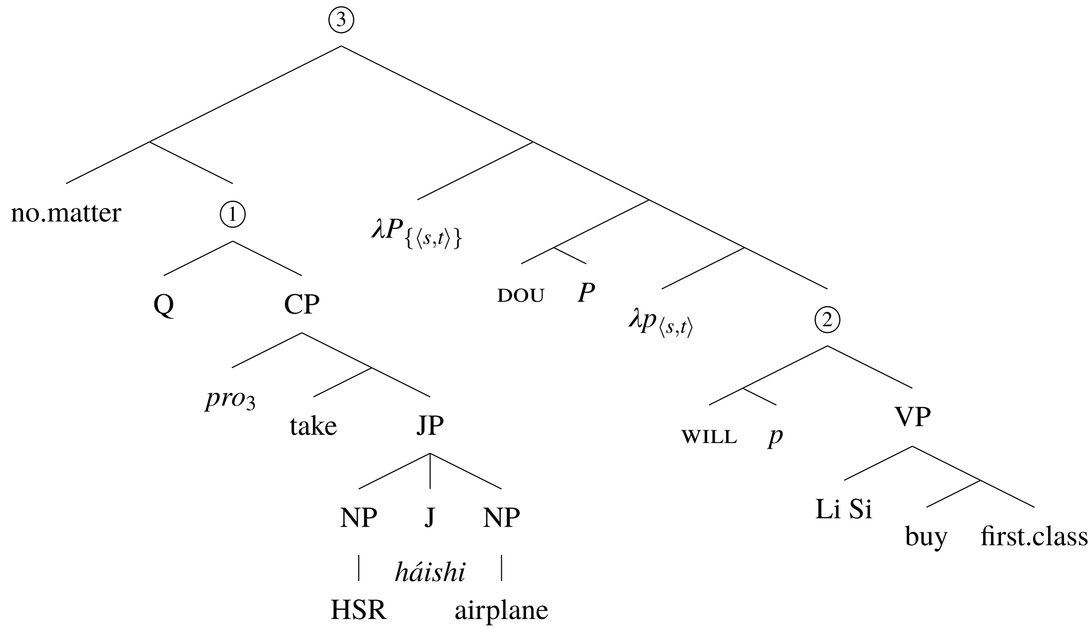
[$_{\text{uncond}}$ Bùguǎn zuò [$_{\text{JP}}$ gāotiě *háishi* fēijī]], Lǐ Sì dōu huì mǎi tóu-děng-cāng.
no.matter sit high.speed.rail haishi airplane Li Si DOU will buy first-class-cabin
'No matter whether traveling by high speed rail or airplane, Li Si will buy a first-class ticket.'

Here I will adopt a version of the semantics for unconditionals from Rawlins 2008a,b, 2013, which was developed for English unconditionals that are also analyzed as embedding questions. In brief, Rawlins proposes that unconditionals semantically serve as conditional clauses in the Lewis 1975 / Kratzer 1981, 1986 / Heim 1982 sense, restricting the domain of a modal in the consequent clause. They are, at the same time, a question which thereby denotes a set of propositions. Each proposition in the question composes pointwise with the consequent, restricting its modal; Rawlins then posits an unpronounced operator that universally quantifies over all of these resulting conditionals, in effect requiring in the case of (42) both that (i) if he takes high speed rail, Li Si travels in first class and (ii) if he flies, Li Si travels in first class.

My formal presentation differs slightly from Rawlins' due to my use of a two-dimensional Alternative Semantics, where he uses a one-dimensional Hamblin semantics, and to better reflect the form of Mandarin unconditionals. Specifically, I propose that *dōu* in unconditionals serves the role of taking the antecedent with a question denotation, i.e. a set of propositions P, and requiring that each $p \in P$ binding the modal restrictor makes the prejacent true. This treatment of *dōu* most closely resembles the proposal in Zhao 2019a,b (see "dou_Q" there) and a suggestion in Li and Law 2016 (p. 219 note 13), and also echoes earlier treatments of *dōu* as a distributivity operator (Lin, 1996, 1998a). I however do not intend for it to extend to other uses of *dōu*, especially the so-called scalar uses.[19]

I assume an LF for example (42) as in (43) below, illustrating the necessary variables and binders in the syntax. This follows the presentation in Rawlins 2008a,b, 2013 and elsewhere where a λ-binder is inserted to link the conditional with a restriction on the modal base; in my proposal here, I simply split this binding into two steps, mediated by *dōu*, with the semantics in (44). The proposition p restricts the modal base of the modal, in this case *huì* 'will,' a variety of epistemic necessity modal (see e.g. Wu and Kuo, 2010; Tsai, 2015b; Wu, 2020).

(43)



(44) $[\![\text{DOU}]\!] = \lambda P_{\{\sigma\}}\ .\ \lambda Q_{\langle\sigma,t\rangle}\ .\ \forall p \in P[Q(p)]$

The 'no matter' expression itself (here, *bùguǎn*) does not contribute a semantics, but takes an interrogative CP complement following Cheng and Huang 1996 and Lin 1996. The unconditional clause (①) includes *háishi* and takes Q at its edge, just as I proposed for alternative questions in section 2.1 above. The result is the question denotation as in (45a). I give the denotation for the node labeled ② in (45b) below. The full result for ③ is as in (45c), utilizing the interpretation for *dōu* in (44) with $\sigma = \langle s,t\rangle$. Note that for all three nodes in (45), the alternative set denotation is the singleton set of the ordinary value. For the intended interpretation, I assume the context resolves $g(3) =$ Li Si.

(45) a. $[\![①]\!]^{o} = \{{}^\wedge\mathbf{HSR}(g(3)), {}^\wedge\,\mathbf{airplane}(g(3))\}$ (type $\{\langle s,t\rangle\}$)

b. $[\![②]\!]^{o} = 1$ iff $\forall w \in \text{Acc} \cap p[({}^\wedge\mathbf{first.class}(\text{Li Si}))(w)]$

[19] See for example discussion of scalar *dōu* in Xiang M. 2008, Liao 2011, Liu 2017a,b, 2019, and Xiang Y. 2020. Notably, none of these works attempt to substantially unify the various uses of *dōu* discussed there with the use of *dōu* in unconditionals.

c. $[\![③]\!]^{o} = 1$ iff $\forall p \in \left\{ \begin{array}{l} {}^\wedge\mathbf{HSR}(g(3)), \\ {}^\wedge\mathbf{airplane}(g(3)) \end{array} \right\} [\forall w \in \text{Acc} \cap p[({}^\wedge\mathbf{first.class}(\text{Li Si}))(w)]]$

The unconditional structures that I have discussed thus far in this section have all been clauses which contribute to the interpretation of the main clause as a conditional clause does, restricting its modal base. As Lin (1996) and subsequent work has shown, there are also structures of the form "(no matter) XP ... *dōu* ..." where the content of XP serves as an argument of the predicate that follows *dōu*. I will refer to these as *argument unconditionals*.

An important property of argument unconditionals is that they can also be of subclausal size, unlike the adjunct unconditionals above. As an argument for this idea, Lin (1996) observes that nominal argument unconditionals (what he calls "*wúlùn*-NPs") appear to syntactically saturate argument positions. Lin's own examples (p. 89) involve *wh*-phrases, but here I reproduce a parallel example involving *háishi* from He 2011, in (46). The argument unconditional itself saturates the subject position, rather than describing an antecedent for a subject pronoun.

(46) **Argument unconditional serves as the subject, rather than anteceding it:** (He, 2011: 81)

[$_{\text{uncond}}$	Wúlùn	Zhāng Sān	*háishi*	Lǐ Sì]	(??tā)	dōu	hěn	cōngmíng.
	no.matter	Zhang San	IDISJ	Li Si	3sg	DOU	very	smart

'Zhang San and Li Si are both smart.'

Despite these facts, Lin (1996) also considers the possibility that argument unconditionals of superficially NP size are underlyingly clausal, with pro-drop and some form of copula drop; for example, for (46) above, underlyingly [*wúlùn pro* ~~COP~~ [ZS IDISJ LS]]. However, He (2011) presents a strong argument against the idea of argument unconditionals having a universally clausal underlying structure. Consider the contrast in (47) below. The argument unconditional in (47) serves as the object for the transitive verb *yuànyì* 'hope,' which must take a clausal/propositional object. (Object argument unconditionals must appear in a position above *dōu*, which my account below will explain; see note 20 below.) The fact that (47) requires an overt *shì* for its clausal argument unconditional serves to show that nominal argument unconditionals as in (46) and (47b) cannot be uniformly interpreted as reduced (copular) clauses. I conclude with He (2011) that there truly are argument unconditionals of both clausal and nominal size.

(47) **Minimal pair demonstrating clausal vs nominal argument unconditionals:** (He, 2011: 80)

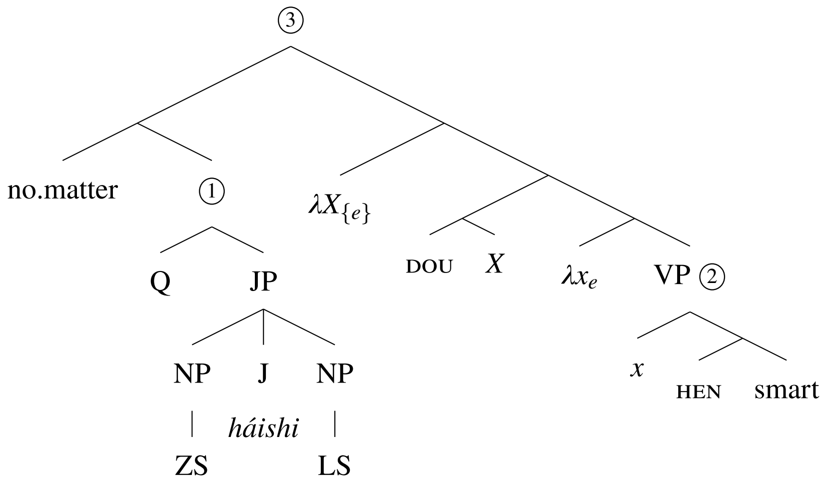
Context: I heard that someone is going to jump off the building and someone asks me who it is.

[$_{\text{uncond}}$	Wúlùn	*(shì)	Zhāng Sān	*háishi*	Lǐ Sì],	wǒ	dōu	bù	yuànyì.
	no.matter	COP	Zhang San	IDISJ	Li Si	1sg	DOU	NEG	hope

'Both that it is Zhang San and that it is Li Si, I do not hope.'

Having established this point on the syntactic size of argument unconditionals, I turn to their semantic analysis. Because argument unconditionals can be of nominal size, unlike the unconditional adjunct clauses above, it does not make sense to describe them as embedded questions. Nonetheless, I suggest that they may be interpreted using the same basic ingredients that I involve for unconditional adjunct clauses above. Concretely, I discuss the interpretation of example (46) above, for which I posit the LF in (48):

(48)



Again I present denotations for key nodes in (49) below. I again assume that the 'no matter' expression (here, *wúlùn*) is semantically vacuous, but here ensures that its complement have an adjoined Q. Although Q generally surfaces at the edge of interrogative CP in my framework, I reiterate that Q is *not* itself a functional head in the CP domain, for instance a variety of a Force head; this allows its adjunction to non-CP categories as well, where necessary. Q here adjoins to JP with disjuncts of NP size. As a result, ① here denotes a set of individuals as its ordinary value (49a). Glossing over the internal composition of the gradable predicate, I assume a denotation for VP (②) as in (49b). DOU in this case takes a set of individuals X that is saturated by the argument unconditional (①) and in turn binds the variable x in the predicate-internal subject position.[20] The resulting interpretation for the entire clause (③) is as in (49c).

[20] To satisfy the thematic requirements of the predicate, as well as the normal requirements of the language's clausal syntax, we may like to think of these two variable-binding dependencies (λX … X, λx … x) as the result of subject movement, possibly

(49) a. $[\![①]\!]^{o} = \{\text{Zhang San}, \text{Li Si}\}$ (type $\{e\}$)

b. $[\![②]\!]^{o} = 1$ iff $\mathbf{smart}(x)$

c. $[\![③]\!]^{o} = 1$ iff $\forall x \in \{\text{Zhang San}, \text{Li Si}\}\ [\mathbf{smart}(x)]]$

Notice that the semantics I give for *dōu* in (44) requires its first argument *X* to have a set denotation as its ordinary value. This serves to semantically enforce that the 'no matter' expression include a Q operator as in (48) above, which in turn ensures that the [uR] feature of *háishi* for Type A speakers is checked in both adjunct and argument unconditionals.

I conclude by noting the existence of certain constructions with *dōu* where *háishi* and *huòzhe* at first glance appear to be interchangeable. Two such examples are in (50).

(50) ***Dōu* structures where *háishi* and *huòzhe* appear to be interchangeable:**

a. Yī-tái cǎidiàn, [JP jiǎ mǎi { *háishi* / *huòzhe* } yǐ mài], dōu shì yī-zhǒng jiàgé.
one-CL color.tv A buy IDISJ / SDISJ B sell DOU COP one-CL price
'A color television, whether A buys it or B sells it, is the same price.' (Ito, 2014: 129)

b. [JP Júzi { *háishi* / *huòzhe* } píngguǒ] dōu xíng.
orange IDISJ SDISJ apple DOU okay
'Oranges and apples are both ok.' (Huang et al., 2009: 242)

However, as a reviewer notes, there are reasons to think that this interchangeability is only apparent. The *háishi* examples here are argument unconditionals of the type discussed above, but without a 'no matter' marker, which is generally optional in Mandarin unconditionals (see (41)). In contrast, the *huòzhe* variants of (50) reflect a distinct construction involving what Xiang (2020) describes as a free choice licensor use of *dōu*. In this latter case, adding a 'no matter' marker is marked[21] and there are various restrictions on the form of the prejacent; see Xiang 2020. I therefore conclude that the *huòzhe...dōu* constructions here are distinct from the (no matter)...*háishi...dōu* structures that I have analyzed here.

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in two steps. Syntactically, the argument unconditional has the external distribution of a NP, and so occupies the high subject position, Spec,TP, in (46/48).

On this account, the need to bind the domain argument of DOU (here, *X*) from above explains the fact that argument unconditionals that correspond to postverbal arguments must necessarily precede *dōu*.

[21] Some prior works claim that 'no matter' markers are available with (certain) *huòzhe...dōu* structures (see e.g. You, 1990; Hou, 2004; Ito, 2014), but they have also been described as highly marked by others (see e.g. Yuan 2021: 77 note 9, Chen 2022: 98). Concretely, Hou (2004: 17) reports 230 "*wúlun...háishi*" constructions as opposed to only 5 "*wúlùn...huòzhe*" examples in the PKU CCL corpus, thus identifying the latter as quite rare, albeit attested. I leave further empirical and theoretical analysis of this possibility for future work.

4.2 Narrow-scope existential *háishi* patterning with existential *wh* (Type B speakers)

I now turn to non-interrogative uses of *háishi* with narrow-scope existential interpretation in certain licensing environments. As noted above, these uses are available only for a subset of speakers, who I call Type B speakers. So as to avoid any risk of potential misunderstanding, I will continue to report these judgements that vary between Type A and Type B speakers with a % judgment mark here below. The goal of this section will be to describe and analyze the behavior of speakers who do allow these uses; that is, those who judge the % examples in this section as fully acceptable with their intended narrow-scope construals as reflected in their translations.[22]

My description in this section draws substantially from previous work by native speaker linguists who report Type B judgments as reflecting their own grammars and that of other speakers they consulted. The most detailed of these studies is Lin Hsin-yin's 2008 master's thesis; much of this almost 200 page work is dedicated to the description of such environments. The striking generalization that she presents is that existential *háishi* is available for these speakers in the same environments that license existential uses of *wh*-phrases in Mandarin. I present some examples here to highlight these parallels.

(51) **Existential *háishi* (Type B) and *wh* under negation:**

a. %Tā méiyǒu mǎi [$_{\text{JP}}$ píngguǒ *háishi* lízi].
3sg NEG.PFV buy apple IDISJ pear
'He/she didn't buy apples or pears.' =(38) (Lin H.-Y., 2008: 1)

b. Tā méiyǒu mǎi *shénme*.
3sg NEG.PFV buy what
'He/she didn't buy anything.' (Cheng, 1984: 102)

(52) **Existential *háishi* (Type B) and *wh* in conditionals:**

a. %Rúguǒ [$_{\text{JP}}$ Lǎo Wáng *háishi* Lǎo Lǐ] lái dehuà, qǐng tōngzhī wǒ.
if old-Wang IDISJ old-Li come COND please notify 1sg
'If Wang or Li comes, please notify me.' (Lin H.-Y., 2008: 141)

b. Rúguǒ *shénme rén* xǐhuān tā, jiù gēn wǒ jiǎng.
if what person like 3sg then with 1sg speak
'If someone likes him/her, then tell me.' (Li, 1992: 136)

[22] As a reviewer notes, the *háishi* examples here may also have wide-scope existential parses with speaker ignorance inferences, even for Type A speakers. I discuss these readings in section 4.3 below.

(53) **Existential *háishi* (Type B) and *wh* under epistemic necessity modals:**

a. %Tā yīdìng jiàn-guò [$_{\text{JP}}$ Zhāng Sān *háishi* Lǐ Sì].
3sg must see-ASP Zhang San IDISJ Li Si
'He/she must have seen Zhang San or Li Si.' (Lin H.-Y., 2008: 80)

b. Tā yīdìng shì bèi *shénme shì* gěi dāngē le.
3sg must SHI PASS what thing make delay LE
'He/she must have been delayed by something.' (Lin J.-W., 1998b: 223)

For the (a) examples with narrow-scope existential *háishi* in (51–53) above, I reproduce examples from Lin Hsin-yin 2008, but structurally parallel examples are reported as grammatical by Hsieh (2004: 89), R. Huang (2010a: 130–131), and Tsai (2015a: 49–50). These works corroborate each other and attest to the robust existence of a population of speakers for whom *háishi* can be used existentially in these environments where existential *wh* are licensed as in (51–53) above.[23] I note as well that all speakers allow the use of the *huòzhe* standard disjunction in the (a) examples for the intended declarative interpretations, which is also predicted on my account, as I discuss below.

Here for concreteness, I adopt the approach to the distribution of existential *wh*-phrases in these licensing environments in Liu and Yang 2021 and demonstrate how it extends to derive the distribution of narrow-scope existential *háishi* in these same environments for Type B speakers. The proposal in Liu and Yang 2021 (also sketched briefly in Liu 2019) builds on much recent work on the analysis of various types of indefinites as in Chierchia 2013 and subsequent work, where these expressions introduce alternatives which must be interpreted by an appropriate covert operator that then has the effect of restricting its distribution. Specifically, Liu and Yang (2021) propose to treat Mandarin *wh*-phrases as existential quantifiers that activate alternatives that correspond to individual atoms in the domain of the *wh*-phrase (in their terms, "singleton domain alternatives"), which must be interpreted by the exhaustification operator O.

I recast Liu and Yang's (2021) assumptions in the following way, to make them compatible with my overall proposal. First, as already introduced above in section 2.1, I follow Ramchand 1997, Beck 2006, Kotek 2019 and others in treating *wh*-phrases as projecting the set of individuals in their domain as their alternative set (just as Liu and Yang require) but with no ordinary defined value; see (54). I instead propose that $\exists_{\text{pass}}$ can be optionally adjoined to the clausal spine — freely, without a featural trigger —

[23] In addition to the environments in (51–53) above, Lin Hsin-yin (2008) reports that existential *háishi* is grammatical for Type B speakers in the following environments, which all license existential *wh*-phrases: in *ma* and A-not-A polar questions (pp. 63–65), under negative adverbs (pp. 65–68, 87), under epistemic possibility modals (pp. 75, 118), under a deontic possibility modal (pp. 80, 103), under non-factive embeddings such as 'think' and 'hope' (pp. 76–78), and with sentence-final *le* (p. 88).

to introduce the disjunctive ordinary value. (I discuss the structural position of $\exists_{\text{pass}}$ below.) Second, I give a syncategorematic entry for O to make explicit that O uses the Roothian alternative set denotation of its sister for its interpretation. O of α requires that its prejacent $[\![\alpha]\!]^{\text{o}}$ be true and that all non-weaker alternatives of the prejacent be false. (55b) specifies that O is resetting.

(54) a. $[\![\text{who}]\!]^{\text{o}}$ undefined b. $[\![\text{who}]\!]^{\text{alt}} = \{x : x \text{ animate}\} = \{\text{ZS, LS, WW, ...}\}$ =(14)

(55) a. $[\![[O\ \alpha]]\!]^{\text{o}} = [\![\alpha]\!]^{\text{o}} \wedge \forall q \in [\![\alpha]\!]^{\text{alt}} [([\![\alpha]\!]^{\text{o}} \not\Rightarrow q) \rightarrow \neg q]$ b. $[\![[O\ \alpha]]\!]^{\text{alt}} = \{[\![[O\ \alpha]]\!]^{\text{o}}\}$

To be clear, I propose the free introduction of $\exists_{\text{pass}}$ and O for all Mandarin Chinese speakers, explaining the general availability of non-interrogative, existential uses of *wh*-phrases in these licensing environments for all speakers. I will highlight the point where I predict the behaviors of Type A and Type B speakers to diverge below. I then extend this analysis to wide-scope existential *háishi* with obligatory speaker ignorance inferences, available for both Type A and Type B speakers, in section 4.3 below.

I first consider the semantics of a basic *wh*-containing clause, without any of the licensing environments for an existential use such as those surveyed above, and discuss what interpretations it may support. As discussed by Cheng (1991), Li (1992), Lin Jo-wang (1998b), and others, (56) may be used as a *wh*-in-situ question, but generally has no declarative use.[24] Based on the denotation for *shéi* 'who' in (54), the denotation for the VP in (56) with the subject interpreted in its predicate-internal position is as in (57). Setting aside the contribution of tense and aspect, the denotation for VP is equivalent to that of TP in this simple case.

(56) [$_{\text{TP}}$ Zhāng Sān xǐhuān *shéi*]
Zhang San like who
i. * 'Zhang San likes someone.'
ii. ✓'Who does Zhang San like?'

(57) a. $[\![\text{VP}]\!]^{\text{o}}$ undefined b. $[\![\text{VP}]\!]^{\text{alt}} = \{{}^\wedge\textbf{like}(\text{ZS}, \text{ZS}),\ {}^\wedge\textbf{like}(\text{ZS}, \text{LS}),\ {}^\wedge\textbf{like}(\text{ZS}, \text{WW}),\ ...\}$

What interpretations does our framework predict (56) to have? First, we may adjoin the Q operator (12) to the clause edge, resulting in a grammatical *wh*-question (58a), deriving the attested reading. Second, we might imagine adjoining the freely available $\exists_{\text{pass}}$ to the VP. The addition of $\exists_{\text{pass}}$ now gives us a defined ordinary value in (58b), but this proposition is not included within the set of alternatives, so the structure violates the principle of Interpretability (3), as I also previewed in (18) above. Finally,

[24] The exception is the possibility of epistemic indefinite uses of *wh*-words, which I discuss in section 4.3. For independent reasons, this indefinite use is very difficult or impossible to access for the structure in (56), as discussed in Liu and Yang 2021: 596–597.

we consider adjoining O to the structure with $\exists_{pass}$ in (58c).[25] O has the "resetting" property (55b), making the result now satisfy Interpretability: the ordinary value will be a member of its alternative set. However, the resulting meaning will be inherently contradictory: because O requires the negation of all non-weaker alternatives of the prejacent, the result requires both that Zhang San like someone and that each alternative of the form "Zhang San likes x" for animate x be false. This is the logic by which Liu and Yang (2021) rule out existential non-interrogative uses of *wh*-phrases in simple clauses such as (56).

(58) **Possible LFs building on (56):**

a. $^{\checkmark}$[Q [... [$_{\text{VP}}$ Zhang San like *who*]]] ⇒ grammatical *wh*-question (56ii)

b. *[$\exists_{pass}$ [$_{\text{VP}}$ Zhang San like *who*]] ⇒ violates Interpretability (3)

i. $[\![[\exists_{\text{pass}}\ \text{VP}]]\!]^{\text{o}} = \exists x\ .\ x \text{ animate} \wedge \textbf{like}(\text{ZS}, x)$

ii. $[\![[\exists_{\text{pass}}\ \text{VP}]]\!]^{\text{alt}} = [\![\text{VP}]\!]^{\text{alt}} = \{\textbf{like}(\text{ZS},\text{ZS}),\ \textbf{like}(\text{ZS},\text{LS}),\ \textbf{like}(\text{ZS},\text{WW}),\ \ldots\}$

c. *[O [$\exists_{pass}$ [$_{\text{VP}}$ Zhang San like *who*]]] ⇒ systematic contradiction

$$\begin{aligned}[\![[O\ [\exists_{\text{pass}}\ \text{VP}]]]\!]^{\text{o}} &= [\![[\exists_{\text{pass}}\ \text{VP}]]\!]^{\text{o}} \wedge \forall q \in [\![[\exists_{\text{pass}}\ \text{VP}]]\!]^{\text{alt}} \left[\left([\![[\exists_{\text{pass}}\ \text{VP}]]\!]^{\text{o}} \not\Rightarrow q\right) \rightarrow \neg q\right] \\ &= (\exists x\ .\ x \text{ animate} \wedge \textbf{like}(\text{ZS}, x)) \wedge \\ &\quad \neg\textbf{like}(\text{ZS},\text{ZS}) \wedge \neg\textbf{like}(\text{ZS},\text{LS}) \wedge \neg\textbf{like}(\text{ZS},\text{WW}) \wedge \ldots\end{aligned}$$

This same logic for the *wh*-containing structure in (56–58) above extends to parallel structures with a *háishi* disjunction in place of the *wh*-phrase, for all Mandarin speakers, thereby explaining the alternative question interpretation for structures such as, literally, "Zhang San likes [Li Si *háishi* Wang Wu]" discussed in section 2.1 above. I also note that, under certain circumstances, these simple *wh*- or *háishi* disjunction-containing structures also allow for wide-scope existential readings with speaker ignorance inferences, the derivation for which I discuss in section 4.3.

I now turn to the licensing contexts, starting with negation. I first schematically consider the addition of negation to the structures in (58) above. First, we continue to allow for an interpretation using Q, leading to a *wh*-in-situ question with negation, in (59a). What differs now is the grammaticality of the parse with both $\exists_{pass}$ and O in (59c), with negation intervening. Downward-entailing operators such as negation lead to a reversal in the entailment relationships between the ordinary value and its alternatives in the scope of O: whereas the prejacent was entailed by each of its alternatives in [$\exists_{pass}$ VP] (58b), the prejacent in [NEG [$\exists_{pass}$ VP]] now entails each of its alternatives in (59b). As a result, the application of O in (59c) will be vacuous, as there are no non-weaker alternatives to negate, but it does serve to reset the

[25] I do not consider adjoining O directly to a *wh*- or J-containing phrase, without first adjoining $\exists_{pass}$, as the semantics of O relies on its sister having a defined ordinary value; see (55).

alternative set denotation, resulting in a meaning that is both contingent and satisfying Interpretability. We therefore predict the existential *wh* interpretation, scoping under negation, to be grammatical.

(59) **Possible LFs with the addition of negation:**

a. $^{\checkmark}$[Q [... NEG ... [$_{\text{VP}}$ Zhang San like *who*]]] ⇒ grammatical *wh*-question

b. *[NEG [$\exists_{\text{pass}}$ [$_{\text{VP}}$ Zhang San like *who*]] ⇒ violates Interpretability (3)

i. $[\![[\text{NEG}\ [\exists_{\text{pass}}\ \text{VP}]]]\!]^{\text{o}} = \nexists x\ .\ x \text{ animate} \wedge \textbf{like}(\text{ZS}, x)$

ii. $[\![[\text{NEG}\ [\exists_{\text{pass}}\ \text{VP}]]]\!]^{\text{alt}} = \{\neg\textbf{like}(\text{ZS}, \text{ZS}),\ \neg\textbf{like}(\text{ZS}, \text{LS}),\ \neg\textbf{like}(\text{ZS}, \text{WW}),\ ...\}$

c. $^{\checkmark}$[*O* [NEG [$\exists_{\text{pass}}$ [$_{\text{VP}}$ Zhang San like *who*]]]] ⇒ grammatical existential *wh*

i. $[\![[O\ [\text{NEG}\ [\exists_{\text{pass}}\ \text{VP}]]]]\!]^{\text{o}} = [\![[\text{NEG}\ [\exists_{\text{pass}}\ \text{VP}]]]\!]^{\text{o}}$ (vacuous)

ii. $[\![[O\ [\text{NEG}\ [\exists_{\text{pass}}\ \text{VP}]]]]\!]^{\text{alt}} = \{[O\ [\text{NEG}\ [\exists_{\text{pass}}\ \text{VP}]]]\}$ (resetting)

The proposal from Liu and Yang 2021 also extends to existential uses of *wh*-phrases under epistemic modals. (60) below presents the results of LFs as in (58) and (59) above, but with a necessity modal $\Box$ reflecting *yīdìng* 'must/certainly' in (53) above. Unlike with negation in (59c) where the result of *O* was vacuous (although with the side effect of resetting the alternative set, to satisfy Interpretability), the application of *O* is not vacuous in (60c). I refer interested readers to Liu and Yang 2021 for discussion of additional licensing environments, including with possibility modals.

(60) **Possible LFs with the addition of necessity modal:**

a. $^{\checkmark}$[Q [... $\Box$ ... [$_{\text{VP}}$ Zhang San like *who*]]] ⇒ grammatical *wh*-question

b. *[$\Box$ [$\exists_{\text{pass}}$ [$_{\text{VP}}$ Zhang San like *who*]] ⇒ violates Interpretability (3)

i. $[\![[\Box\ [\exists_{\text{pass}}\ \text{VP}]]]\!]^{\text{o}} = \Box\,(\exists x\ .\ x \text{ animate} \wedge \textbf{like}(\text{ZS}, x))$

ii. $[\![[\Box\ [\exists_{\text{pass}}\ \text{VP}]]]\!]^{\text{alt}} = \{\Box\,\textbf{like}(\text{ZS}, \text{ZS}),\ \Box\,\textbf{like}(\text{ZS}, \text{LS}),\ \Box\,\textbf{like}(\text{ZS}, \text{WW}),\ ...\}$

c. $^{\checkmark}$[*O* [$\Box$ [$\exists_{\text{pass}}$ [$_{\text{VP}}$ Zhang San like *who*]]]] ⇒ grammatical existential *wh*

$$[\![[O\ [\Box\ [\exists_{\text{pass}}\ \text{VP}]]]]\!]^{\text{o}} = \Box\,(\exists x\ .\ x \text{ animate} \wedge \textbf{like}(\text{ZS}, x)) \wedge$$
$$\neg\Box\,\textbf{like}(\text{ZS}, \text{ZS}) \wedge \neg\Box\,\textbf{like}(\text{ZS}, \text{LS}) \wedge \neg\Box\,\textbf{like}(\text{ZS}, \text{WW}) \wedge ...$$

As per the discussion of the basic case in (56–58) above, the considerations of the compositional semantics for the LFs in (59) and (60) also extend to variants of these structures with *háishi* disjunctions in place of the *wh*-phrases. That is, based on the considerations of the compositional semantics alone, we predict a *háishi* disjunction in the scope of a downward-entailing operator or a modal to be interpretable both as an alternative question and as a declarative with boolean disjunction scoping under the licensing operator. However, as we have seen, the latter possibility is only available for Type B speakers. As noted above, I propose that the interrogative disjunctor *háishi* bears a [uR] feature for Type A speakers, requiring its association with a root-level operator such as Q (or another operator, for wide-scope ignorance readings; see section 4.3); this blocks LFs of the form in (59c) and (60c) with *háishi* disjunction as the alternative source. In contrast, Type B speakers do not restrict the operator involved in the evaluation of *háishi*'s alternatives, allowing for these parses.

I highlight an additional detail regarding these existential uses of *wh*-phrases and *háishi* under negation. Lin Hsin-yin (2008) reports that the perfective negation *méi(yǒu)* (51a) as well as the negator *búshì* allow for the non-interrogative, existential use of *háishi* for Type B speakers, but the simple negation *bù* does not; see (61a). Notably, this parallels a contrast between *búshi* (and *méi(yǒu)*; see (51)) versus *bù* in the availability of existential *wh*-phrases as well; see (61).[26] Note that the judgment marks in (61) are all for the intended, narrow-scope existential readings.

(61) **Existential *wh* and *háishi* (Type B) under high vs low negations:** (Lin, 2008: 51–53)

a. Tā { %búshì / *bù } xǐhuān [$_{JP}$ Zhāng Sān *háishi* Lǐ Sì].
3sg NEG / NEG like Zhang San IDISJ Li Si
'He/she doesn't like Zhang San nor Li Si.'

b. Tā { ✓búshì / *bù } tǎoyàn *shéi*.
3sg NEG / NEG hate who
'He/she doesn't hate anyone.'

I argue that this difference reflects a restriction on the structural positions where $\exists_{pass}$ can be freely adjoined. The negator *bù* is structurally lower than *méiyǒu* and *búshì* (Huang, 1988b; Yeh, 1992; Hsieh, 1996; Erlewine, 2017). I propose that $\exists_{pass}$ cannot be adjoined under *bù*,[27] accounting for this contrast between the negators and their parallel between the licensing of existential *wh*-phrases and *háishi* disjunction for Type B speakers. The possibility of accounting for this contrast by restricting the syntactic positions for $\exists_{pass}$ constitutes another advantage of my implementation of Liu and Yang's (2021) theory over the original, where *wh*-phrases are treated as inherently denoting existential quantifiers as their ordinary values.

Beyond this lower bound on the position of $\exists_{pass}$, however, its adjunction position is flexible as long as the end result is meaningful and satisfies Interpretability. In particular, in environments where there

[26] The degradedness of *bù* as a licensor of existential *wh*-phrases is reflected in some notes in prior work as well. See for instance Li 1992: 150 note 3 and Lin J.-W. 2004: 460 note 8.

[27] Lin (2008: 54–56) shows that *bù* can license existential *háishi* in an embedded clause. This too is explained by my account, as $\exists_{pass}$ can then be adjoined within the lower clause itself.

are multiple potential licensors, I suggest that $\exists_{\text{pass}}$ may be adjoined at different heights; this follows the approach described by Li and Law (2016: 221–224) for scope facts described by in Lin 2004. In particular, Li and Law (2016: 222–224) show that the span between the alternative source and the position of the existential operator (here, $\exists_{\text{reset}}$) is susceptible to focus intervention effects with *zhǐyǒu* 'only' (see their examples (53) vs (57) and (58) vs (59)), just as *wh*-questions and *háishi* alternative questions and the scope-taking of *huòzhe* disjunction are affected by such intervention effects as well (see §3.2). These intervention effect facts further support my approach here, where the existential force of existential *wh*-phrases and *háishi* disjunction for Type B speakers is introduced by an adjoined sentential operator that quantifies over projected Roothian alternatives.

Finally, I note that in all of the constructions discussed in this section, where Type B speakers allow for a narrow-scope existential use of *háishi*, the standard disjunctor *huòzhe* can be used by all speakers to express the same intended reading. This too is predicted by my account. For all speakers, the standard disjunction *huòzhe* bears the feature [u$\exists$] which must be checked by either $\exists_{\text{pass}}$ or $\exists_{\text{reset}}$. The parses discussed in this section involving $\exists_{\text{pass}}$ are also possible with *huòzhe*, checking its [u$\exists$] feature.[28]

4.3 Wide-scope existential *háishi* with speaker ignorance/irrelevance

The final class of non-interrogative uses of *háishi* is as a wide-scope existential, in situations where the speaker cannot and need not distinguish between the disjuncts for the purposes of the discourse. I reproduce two illustrative examples from Yuan 2021 in (62) below. Example (62a) is a test item from an acceptability judgement experiment that Yuan conducted, which I discuss below, whereas example (62b) is a naturally occurring corpus example. Importantly, these wide-scope existential uses of *háishi* are possible for *all* Mandarin speakers, unlike the narrow-scope existentials under certain licensing operators described in the previous section (§4.2).

(62) **Existential *háishi* with speaker ignorance/irrelevance inferences:** (Yuan, 2021: 71, 73)

a. Xiǎo-Lín zhōu-mò qù-le [JP Xiānggǎng *háishi* Àomén] gòuwù, mǎi-le
little-Lin week-end go-PFV Hong Kong IDISJ Macao shop buy-PFV
yī-dà-duī huàzhuāngpǐn.
one-large-pile cosmetics
'Lin went shopping in Hong Kong or Macao over the weekend and bought a lot of cosmetics.'

[28] I note that the interchangeability of *háishi* and *huòzhe* in these licensing environments for Type B speakers constitutes an argument against a hypothetical alternative account where there is a single disjunctor morpheme in the lexicon, J, whose realization is determined post-syntactically based on the choice of operator that it associates with.

b. Máo zhǔxí zài [JP 1943-nián *háishi* 1944-nián] gěi tā xiě-guò yī-ge
Mao chairman LOC 1943-year IDISJ 1944-year to 3sg write-EXP one-CL

jiǎngzhuàng.
certificate

'Chairman Mao had written him a certificate of merit in 1943 or 1944.'

Experimental results reported in Yuan 2021 provide strong evidence for the existence of a population of speakers with Type A judgments as I have described here: that is, allowing for universal *háishi* as in section 4.1 and the wide-scope existential uses with speaker ignorance as in (62), but *not* the narrow-scope existential uses of section 4.2. Yuan reports on an acceptability rating experiment conducted with 175 speakers in mainland China, with Latin square design. She reports that example items of the form in (62a) were judged as relatively acceptable. Crucially, Yuan also included conditions to test for the acceptability of narrow-scope existential *háishi* in a number of the licensing environments for Type B speakers discussed in section 4.2 above; she reports that all of these were judged unnatural by her participants. Throughout this section, I will discuss additional findings from her experiment as informative regarding the Type A profile of judgments. I will also summarize the results of her experiment, including its experimental conditions and their average ratings, later in this section.

Similar to the discussion of other non-interrogative uses of *háishi* above, my proposal here will build on existing work on a parallel use of *wh*-words in the language. *Wh*-words in Mandarin Chinese can be used as so-called epistemic indefinites, which introduce obligatory speaker ignorance inferences (see also Alonso-Ovalle and Menéndez-Benito, 2013, 2015), subject to certain pragmatic conditions which I discuss below. Consider example (63). The first sentence in (63) comes from the 1961 novel *Hóngyán* [*Red Crag*], with the infelicitous continuation reflecting an obligatory inference that the speaker is unable to identify the referent of the indefinite *shéi*.[29]

(63) **Epistemic indefinite *shéi* 'who' with speaker ignorance:** (Liu and Yang, 2021: 587)

Gǒuxióng zhèngzài dī-shēng hé *shéi* jiǎnghuà. (#Wǒ kàn-de-qīngqīngchǔchǔ,
Gouxiong PROG low-voice with who speak 1sg see-DE-clear

nà-ge rén jiù shì Xīngxīng.)
that-CL person then COP Xingxing

'Gouxiong is talking to someone quietly.'

Intended continuation: 'I could see clearly; that person was Xingxing.'

I will describe the wide-scope existential *háishi* as an epistemic disjunction, and analyze it as parallel to the epistemic indefinite use of *wh*-words in Mandarin. Recall from the preceding section that *wh*- and *háishi*-disjunction-containing clauses by themselves do not lead to grammatical declarative LF structures, even with the availability of freely adjoining $\exists_{\text{pass}}$ (16) and O (55). Following the analysis of epistemic indefinite *wh*-words in Liu and Yang 2021 — and in turn following much previous work on the derivation of ignorance inferences (see e.g. Alonso-Ovalle and Menéndez-Benito, 2010; Chierchia, 2013; Meyer, 2013) — I adopt the idea that there is a covert speaker-oriented doxastic necessity modal (here notated $\Box_s$) in the clause periphery of assertions, reflecting the sincerity conditions on their felicitous use. See the LF based on example (63) above in (64) below. With the addition of an existential operator with the semantics of $\exists_{\text{pass}}$ (but here labeled $\exists_{\text{R}}$, for reasons described below) and O, we successfully derive an Interpretable and contingent result, which includes the desired ignorance inferences.

(64) **LF for (63) with covert necessity modal:**

[O [$\Box_s$ [$\exists_{\text{R}}$ [$_{\text{TP}}$ Gouxiong talks to *who*]]]]

$\Rightarrow \Box_s\, (\exists x\, .\, x \text{ animate} \land \textbf{talk}(\text{GX}, x)) \land \neg\, \Box_s\, \textbf{talk}(\text{GX}, \text{ZS}) \land \neg\, \Box_s\, \textbf{talk}(\text{GX}, \text{LS}) \land \ldots$

This approach in (64) extends directly to the derivation of wide-scope existential uses of *háishi* with their ignorance inferences as well. The obligatory speaker ignorance inferences also correctly predict that epistemic disjunction *háishi* is less acceptable in first-person reports Yuan (2021: 75), just as has also been reported with *wh* epistemic indefinites (Chen, 2018: 148–149).

I comment now on the special existential operator labeled $\exists_{\text{R}}$ which I proposed for configurations such as in (64). Recall that this epistemic disjunction of *háishi* is available for both Type A and Type B speakers, unlike the narrow-scope existential uses discussed in section 4.2 above. On my account of the Type A versus Type B distinction, *háishi* for Type A speakers has a syntactic feature [uR], whose checking requirement serves to restrict its distribution. Concretely, I propose that $\exists_{\text{R}}$ can also check the [uR] feature, but $\exists_{\text{pass}}$ cannot. Combined with the restriction that $\exists_{\text{R}}$ can only be adjoined in the periphery of root clauses, we derive the restriction that grammatical existential uses of *háishi* for Type A speakers must take wide-scope and introduce speaker ignorance inferences, blocking the narrow scope adjunction of an existential operator that would lead to licensing in the environments that allow for narrow-scope existential *háishi* for Type B speakers. I summarize the salient properties of $\exists_{\text{R}}$ in (65):

[29] There are also additional modal indefinite uses of *shénme* 'what' as ranging over kinds or degrees (Liao, 2011; Huang, 2013; Chierchia and Liao, 2015; Chen, 2017, 2021), which lead to various inferences of "agent indifference" or "insignificance," but these are specific to the *wh*-word *shénme*. See (69) below for examples. Example (63) here illustrates the intended epistemic indefinite use with *shéi* 'who,' which simply ranges over animate individuals.

(65) **An additional existential operator $\exists_R$, for wide-scope existential *háishi*:**

a. Semantics: $[\![\exists_R]\!] \equiv [\![\exists_{pass}]\!]$ (16)

b. Syntax: Can be adjoined freely in the periphery of root clauses, but nowhere else. Checks [uR] on *háishi* for Type A speakers (39).

I should immediately clarify that $\exists_R$ may also be adjoined to embedded clauses under attitude predicates, which have "embedded root" status (see e.g. Heycock, 2006), with a corresponding shift in doxastic base. Here I discuss the two variants of example (66) below, which embeds a truncated form of example (62a) above.

(66) **Existential *háishi* with ignorance in attitude reports:**

{ Xiǎo-Míng / Wǒ } júede [$_{CP}$ Xiǎo-Lín zhōu-mò qù-le [$_{JP}$ Xiānggǎng *háishi* Àomén]
little-Ming 1sg think little-Lin week-end go-PFV Hong Kong IDISJ Macao

gòuwù].
shop

'{Ming thinks / I think} [that Lin went shopping in Hong Kong or Macao over the weekend].'

I first discuss the third-person variant of example (66). On one reading, the speaker asserts that [Ming believes Lin went shopping in HK] or [Ming believes Lin went shopping in Macao], but the speaker doesn't know which. This is the wide-scope existential reading with speaker ignorance derived straightforwardly from the LF in (67a), modeled after the logic of (64) above. Note that I here represent 'Ming think' of the matrix clause simply as a doxastic necessity modal over Ming's belief worlds, $\Box_{Ming}$. The sentence is also valid as a report that Ming believes Lin went shopping in HK or Macao but Ming is unsure which. This parse requires adjunction of $\exists_R$ in the embedded clause periphery as in (67b). (Recall that *O* can be adjoined freely where it is semantically motivated.)

(67) **Simplified LFs for (66) with third-person subject:**

a. [$_{CP}$ ***O*** [$\Box_s$ [$\boldsymbol{\exists_R}$ [$_{TP}$ $\Box_{Ming}$ [$_{CP}$ [$_{TP}$ Lin went shopping in [$_{JP}$ HK *háishi* Macao]]]]]]]

$\Rightarrow \Box_s \big(\Box_{Ming}\, \mathbf{shop}(L, HK) \vee \Box_{Ming}\, \mathbf{shop}(L, M)\big) \wedge$
$\neg \Box_s \Box_{Ming}\, \mathbf{shop}(L, HK) \wedge \neg \Box_s \Box_{Ming}\, \mathbf{shop}(L, M)$

b. [$_{CP}$ $\Box_s$ [***O*** [$_{TP}$ $\Box_{Ming}$ [$_{CP}$ $\boldsymbol{\exists_R}$ [$_{TP}$ Lin went shopping in [$_{JP}$ HK *háishi* Macao]]]]]]

$\Rightarrow \Box_s \big(\Box_{Ming}[\mathbf{shop}(L, HK) \vee \mathbf{shop}(L, M)] \wedge \neg \Box_{Ming}\, \mathbf{shop}(L, HK) \wedge \neg \Box_{Ming}\, \mathbf{shop}(L, M)\big)$

The first-person variant of example (66), which is judged to be natural, provides an even clearer argument for the possibility of embedded clause attachment for $\exists_{\text{R}}$. The LFs in (68) below are based on those in (67), but with the matrix clause expressing the speaker's own belief with $\Box_s$, rather than $\Box_{\text{Ming}}$. Notice that the parse in (68a) with matrix attachment of $\exists_{\text{R}}$ leads to a contradictory result: (the speaker believes that) [the speaker believes that Lin went to HK] or [the speaker believes that Lin went to Macao], but simultaneously believes neither of these claims. In contrast, the LF in (68b) with $\exists_{\text{R}}$ adjunction in the embedded clause periphery, within the scope of the matrix 'I think,' correctly derives the attested interpretation.

(68) **Simplified LFs for (66) with first-person subject:**

a. $^{*}[_{\text{CP}}\ \boldsymbol{O}\ [\ \Box_s\ [\ \boldsymbol{\exists_{\text{R}}}\ [_{\text{TP}}\ \Box_s\ [_{\text{CP}}\ [_{\text{TP}}$ Lin went shopping in $[_{\text{JP}}$ HK *háishi* Macao]]]]]]]
$\Rightarrow \Box_s\ (\Box_s\ \textbf{shop}(\text{L, HK}) \vee \Box_s\ \textbf{shop}(\text{L, M})) \wedge \neg \Box_s \Box_s\ \textbf{shop}(\text{L, HK}) \wedge \neg \Box_s \Box_s\ \textbf{shop}(\text{L, M})$

b. $[_{\text{CP}}\ \Box_s\ [\ \boldsymbol{O}\ [_{\text{TP}}\ \Box_s\ [_{\text{CP}}\ \boldsymbol{\exists_{\text{R}}}\ [_{\text{TP}}$ Lin went shopping in $[_{\text{JP}}$ HK *háishi* Macao]]]]]]
$\Rightarrow \Box_s\ (\Box_s\ [\textbf{shop}(\text{L, HK}) \vee \textbf{shop}(\text{L, M})] \wedge \neg \Box_s\ \textbf{shop}(\text{L, HK}) \wedge \neg \Box_s\ \textbf{shop}(\text{L, M}))$

My proposal for existential *háishi* with ignorance inferences also extends to cases that include a disjunct with a *shénme* 'what' modifier, as in (69) below. Example (69a) is reproduced here with Cheng's (1984) own translation, where *háishi shénme de* corresponds to 'or something like that,' and I have modeled my translation for (69b) accordingly. Yuan (2021) gives example (69b) as a representative item for her "*háishi* + question word" experimental condition, which she reports as judged to be reasonably acceptable by her participants who, as noted above, are Type A speakers.

(69) ***Háishi* + *shénme (de)* as 'or something like that':**

a. Tā xiǎng chī diǎnr $[_{\text{JP}}$ xiāngjiāo, mùguā, *háishi* shénme de]. (Cheng, 1984: 109)
3sg want eat some banana papaya IDISJ what DE
'He/she wants to eat some banana or papaya or something like that.'

b. Xiǎo-Lín zhōu-mò qù-le $[_{\text{JP}}$ Xiānggǎng *háishi* shénme dìfāng]. (Yuan, 2021: 71)
little-Lin week-end go-PFV Hong Kong IDISJ what place
'Lin went to Hong Kong or someplace like that.'

Based on my proposal for J in (4), JP in (69a) will have no defined ordinary value but an alternative set denotation that includes banana and papaya, as well as the alternatives introduced by *shénme de*.[30]

[30] I assume that *shénme* in these examples are kind-denoting modifiers (see e.g. Huang 2013; Chierchia and Liao 2015; Chen 2017

The logic that derives the wide-scope existential semantics with speaker ignorance inference in (64) above then also applies here, with the only difference being that the full set of relevant alternatives is left unspecified, hence signaling further uncertainty and/or irrelevance.

It is important to note that epistemic indefinite *wh*-words and the parallel, epistemic disjunction *háishi* are subject to a pragmatic constraint, such that the ignorance with respect to the referent of the *wh*-word or *háishi* disjunction must not detract from the "main point" of the utterance (see e.g. Yuan, 2021: 75 note 6). In other words, identifying the exact referent must be unnecessary for the current purposes of the conversation. This is illustrated well in the examples we've considered throughout this section. For instance, returning to the naturally occurring *Chairman Mao* example in (62b) above, its main point is that the protagonist was awarded a certificate by Chairman Mao; it is not important to identify the exact date when this occurred.

Yuan's (2021) experimental study indirectly addresses this issue by testing shorter and longer variants of certain examples, which change the pragmatic status of the disjunction. For example, the experiment included both the longer *shopping* example in (70a), repeated from (62a) above, as well as its truncated form in (70b). Yuan reports that examples of the form in (70a) were judged to be relatively natural, as opposed to their corresponding shorter forms as in (70b) which were rated lower, and which I mark ??. Evaluated without context, the continuation in (70a) supports an interpretation that Lin's shopping spree is the main point, with the exact destination becoming part of the "background" (Yuan, 2021: 75 note 6), whereas the shorter form in (70b) does not support such an interpretation.

(70) **Testing the effect of elaborations that shift the main point:** (Yuan, 2021: 71)

a. Xiǎo-Lín zhōu-mò qù-le [JP Xiānggǎng *háishi* Àomén] gòuwù, mǎi-le
little-Lin week-end go-PFV Hong Kong IDISJ Macao shop buy-PFV
yī-dà-duī huàzhuāngpǐn.
one-large-pile cosmetics
'Lin went shopping in Hong Kong or Macao over the weekend and bought a lot of cosmetics.'

b. ??Xiǎo-Lín zhōu-mò qù-le [JP Xiānggǎng *háishi* Àomén].
little-Lin week-end go-PFV Hong Kong IDISJ Macao
'Lin went to Hong Kong or Macao over the weekend.'

and note 29 above), rather than the regular 'what' with individual (type *e*) alternatives, and that their alternative set denotations are contextually restricted. This accords with prior descriptions such as Liu Binmei's (2009: 371–372) note that the addition of *shénme de* to a list "has an interpersonal hedging function... to let the hearer infer other similar examples, and also could signal to downplay the importance of what has been said."

Relatedly, various authors have observed that epistemic indefinites in Mandarin must avoid prosodic prominence. Concretely, this restriction explains the observation that *wh*-words in utterance-final object positions with intended epistemic indefinite use require a prosodic shift to destress the *wh*-word (Pan 2007: ch. 4, 2019b, Dong 2009: ch. 3), or may simply be judged as less acceptable, due to this being the default position for nuclear stress placement and focal construal (see e.g. Liu and Yang 2021: 597, Shi and Pan 2023). We can understand this observation as related to the need of epistemic indefinite *wh*-words and similar uses of *háishi* disjunction to avoid being part of the main point of the utterance; see also Simons et al. 2010, Tonhauser 2012, and Koev 2018 on the relationship between focus and "main point" status.

Here I will leave the full description and analysis of the nature of this "irrelevance" requirement for future work. Nevertheless, the recognition of this requirement is important for better understanding how the distinction between *háishi* vs *huòzhe* disjunctors is often presented in the literature. An effect of this constraint is that many simple sentences with a *háishi* disjunction, especially in utterance-final object position, are often judged to be unambiguously interrogative and thus reported as such. This in turn contributes to the initial appearance of a one-to-one correspondence between disjunctor form (*háishi* vs *huòzhe*) and interrogative versus declarative function in many simple examples (including (1)) and in many prior descriptions.

Finally, I propose to extend the proposal in this section to instances of existential *háishi* with ignorance inferences in the presence of *hǎoxiàng* 'seemingly.' I reproduce two such examples from Yuan 2021.[31] Example (71a) is a corpus example. Yuan gives (71b) as representative of her experimental items of existential *háishi* in the presence of *hǎoxiàng*, which she reports as being relatively acceptable for her (Type A) participants.[32]

(71) **Existential *háishi* with speaker ignorance, with *hǎoxiàng* 'seemingly':** (Yuan, 2021: 71,73)

a. Bù hǎo ya, gébì xiǎo-gē hǎoxiàng bǎ tā-de wifi [JP gěi-guān-le
NEG good PRT neighbor little-brother seemingly BA 3SG-POSS wifi make-lock-PFV
háishi gěi-yǐncáng-le].
IDISJ make-hide-PFV
'Oh no, the guy next door seems to have locked or hidden his wifi.'

[32] Lin Hsin-yin (2008) also reports the availability of existential *háishi* in sentences with *hǎoxiàng*, for what I call Type B speakers.
[32] I translate sentences with *hǎoxiàng* using English 'seem,' but *hǎoxiàng* itself is an adverb rather than a raising verb; see e.g. Li 1985: 229–230, 1990: 122–123.

b. Xiǎo-Lín hǎoxiàng zhōu-mò qù-le [$_{JP}$ Xiānggǎng *háishi* Àomén].
little-Lin seemingly week-end go-PFV Hong Kong IDISJ Macao
'Lin seems to have gone to Hong Kong or Macao over the weekend.'

Loosely following the analysis of evidential markers as illocutionary modifiers in Faller 2002, I suggest that *hǎoxiàng* shifts the norm of assertion to express what must be true if we adopt a particular source of evidence as credible. Given my adoption of a covert doxastic necessity modal ($\Box_s$) at the root of assertions, I then technically treat *hǎoxiàng* as licensing and reflecting the addition of evidence *ev* to the modal base of that covert modal (from $\Box_s$ to $\Box_{s+ev}$, as in (72) below); *hǎoxiàng* is then not itself a modal operator.[33] See (72) below, where I strike out *hǎoxiàng* to indicate its semantic vacuity. This approach correctly predicts the availability of wide-scope existential *háishi* in the presence of *hǎoxiàng* for all Mandarin speakers. The result as in (72) reflects that the speaker, taking this piece of evidence *ev* to be credible, is certain that the disjunctive claim is true, but still cannot determine which disjunct is true, even with the additional evidence considered.

(72) **LF for (71b):**
[$_{CP}$ ***O*** [$\Box_{s+ev}$ [$_{TP}$ Lin ~~*hǎoxiàng*~~ went shopping in [$_{JP}$ HK *háishi* Macao]]]]
$\Rightarrow \Box_{s+ev}\,(\mathbf{shop}(L, HK) \vee \mathbf{shop}(L, M)) \wedge \neg \Box_{s+ev}\,\mathbf{shop}(L, HK) \wedge \neg \Box_{s+ev}\,\mathbf{shop}(L, M)$

I conclude this section by further describing the experimental results in Yuan 2021 as particularly informative regarding the difference in judgments between Type A and Type B speakers. Yuan reports on an acceptability judgement experiment where 175 participants in mainland China rated examples (presented in a Latin square design) on a five-point Likert scale. Yuan concludes that the conditions

[33] Some prior works treat *hǎoxiàng* as an epistemic possibility modal (see e.g. Huang, 2018), whereas my proposal here treats its use as reflecting a variety of necessity operator. Support for my description comes from the behavior of utterances of the form "*hǎoxiàng p*; *hǎoxiàng* $\neg p$." Speakers report that examples such as (i) below without the parenthetical material are possible but suggest that the speaker is switching between consideration of two different pieces of evidence (for instance, the appearance of the sky versus the weather forecast). Contextually specifying that only one particular source of evidence is considered, such as with the parenthetical in (i), then leads to a clear judgement of the utterance as anomolous. This differs from the behavior we would expect of an epistemic possibility modal. I especially thank Tom Grano and Ka-Fai Yip (p.c.) for discussion of this issue.

(i) (#Zhǐ kàn tiān-sè,) Jīntiān hǎoxiàng huì xià yǔ, yòu hǎoxiàng bù huì xià yǔ.
only see sky-color today seemingly will fall rain again seemingly NEG will fall rain
'(#Judging based only on the color of the sky,) it seems that it will rain today, and yet again, it seems that it will not.'

On my account here, the use of *hǎoxiàng* suggests via Gricean implicature (see also Faller, 2012) that the speaker's doxastic state alone is insufficient to assert the prejacent *p*. This explains the intuition that *hǎoxiàng p* is logically weaker than *p* and hence *hǎoxiàng* may resemble an epistemic possibility modal.

My treatment here of the relationship between covert $\Box_s$ and *hǎoxiàng* is, in a sense, the opposite of a suggestion made by Tsai (2015b: 50) that wide-scope existential uses of *háishi* may reflect the presence of a high, covert variant of *hǎoxiàng*.

that tested for the acceptability of existential *háishi* can be split into two groups — those that make existential *háishi* relatively natural for her participants, with average ratings between 3 and 4 (73a), and those that scored an average rating of less than 2.5, which Yuan then describes as unnatural (73b) — with a statistically significant difference between them. I reproduce the average ratings for each condition in (73).[34]

(73) **Summary of acceptability judgement ratings from Yuan 2021:**

a. "Long clause" (62/70a): 3.41; with a *wh*-word (69): 3.66; *hǎoxiàng* 'seemingly' (71): 3.47

b. "Simple declarative" (70b): 2.31; epistemic modal: 2.45; deontic modal: 2.44; conditional clause: 2.49; negation: 2.31; polar question: 2.21

These results attest to the existence of a broad group of speakers (Type A) who accept the existential *háishi* with ignorance inferences — as long as the sentence is not too short, in order to avoid uncertainty regarding the "main point" of the utterance, as illustrated in (70) above — and especially with a *wh*-word and with *hǎoxiàng* 'seemingly.' However, these speakers do *not* allow for narrow-scope existential *háishi* in the licensing environments discussed in section 4.2, thus distinguishing themselves from Type B speakers. My proposal in this section derives precisely this distribution by proposing one additional existential operator $\exists_{\mathrm{R}}$ (65) which can satisfy the syntactic requirement of *háishi* for Type A speakers ([uR]), and which is limited to the clause periphery and with the semantics of $\exists_{\mathrm{pass}}$, thereby leading necessarily to wide-scope interpretation with ignorance inferences.

5 Conclusion

In this paper I investigated the distributions and interpretations of the two disjunctors in Mandarin Chinese, *háishi* and *huòzhe*. In the most basic cases, the use of *háishi* necessarily results in an alternative question whereas the use of *huòzhe* results in a disjunctive proposition. Following Haspelmath 2007 and Mauri 2008, this exemplifies an interrogative versus standard disjunctor contrast, and I adopt these terms. However, I have shown here that the distributions of use and interpretations for the two disjunctors is much more complicated that this first description suggests.

My account here offers the first compositional semantics for the Mandarin disjunctors that adequately accounts for restrictions on the scope-taking of both *háishi* and *huòzhe*, which I show to be deeply parallel to one another, as well as the environments that allow for non-interrogative uses of the interrogative

[34] Yuan (2021) did not test constructions that support universal readings of *háishi*, as in section 4.1, most likely due to their judgments not being so variable and controversial as those for existential uses of *háishi* are.

disjunctor *háishi*. Adopting the two-dimensional Alternative Semantics framework of Rooth 1985, 1992 and subsequent work, I propose that both disjunctors project their disjuncts as a set of alternatives in the Roothian alternative set dimension (following Beck and Kim, 2006), which must then be interpreted by an operator for question formation or existential closure, depending on the use. Along the way, I've argued against prior proposals for the Mandarin disjunctors that posit covert scope-taking movement (e.g. Huang 1982: 276 for *háishi* and Erlewine 2014: p. 223 note 5 for *huòzhe*) or where *háishi* but not *huòzhe* must take disjuncts of clausal size (J. Huang et al. 2009: 250, R. Huang 2009, 2010a,b; see Appendix A).

I argue that the adequate description of the nature of the distinction between the two disjunctors requires reference to both syntactic and semantic tools. First, I claim that the standard disjunction *huòzhe* must always cooccur with a corresponding existential operator, which I enforce through syntactic feature-checking. I then propose that the interrogative disjunctor *háishi* may or may not syntactically enforce its cooccurrence with a root-level operator such as Q, corresponding to the distinct behaviors of two broad categories of speakers. For one group of speakers (Type A; see especially Yuan 2021), the interrogative disjunctor *háishi* syntactically requires association with a question operator Q or a wide-scope existential operator that leads to ignorance inferences, while another group of speakers (Type B; see especially Lin Hsin-yin 2008) allow *háishi* to be used non-interrogatively in the same environments that *wh*-phrases do. My analysis for this full set of facts, especially the sensitivity to focus intervention effects and the parallels to non-interrogative *wh*-phrases for Type B speakers, builds on the treatment of *wh*-phrases as projecting Roothian alternatives as in Beck 2006 and Kotek 2019, as well as my Interpretability constraint (3) for the two-dimensional Rooth-Hamblin Alternative Semantics.

Similar lexical distinctions between interrogative and standard disjunction have been noted in a range of languages of the world, of different regions and language families (Haspelmath, 2007; Mauri, 2008). My investigation into Mandarin Chinese *háishi* versus *huòzhe* here offers an example both of the types of empirical phenomena to consider and the potential analytic options for exactly how such a distinction should be described. Concretely, I advocate for investigations into environments that support the exceptional, non-interrogative use of an "interrogative disjunctor" — and potentially, exceptional circumstances under which the "standard disjunctor" is used to form an alternative question — as particularly informative.

A preliminary look shows, perhaps unsurprisingly, that different languages indeed vary in the ranges of use for their two disjunctors in different environments. Consider for example the Vietnamese and Finnish conditional clauses in (74–75) below. Here, we see that Vietnamese patterns with Mandarin Type B (52a) in allowing a non-interrogative use of the interrogative disjunctor inside the question, whereas

Finnish patterns with Mandarin Type A in not allowing such use.[35]

(74) **Vietnamese conditional clause licenses non-interrogative, existential *hay* IDISJ:**

[$_{\text{cond}}$	Nếu	[$_{\text{JP}}$	Minh	{	✓*hoặc*	/	✓*hay*	}	Kim]	gọi	đến	]	thì	bảo	là	tôi	đang	họp.
	if		Minh		SDISJ	/	IDISJ		Kim	call	come		then	say	that	1sg	PROG	meeting

‘If Minh or Kim calls, say that I’m in a meeting.’ (Anne Nguyen, p.c.)

(75) **Finnish conditional clause does not license non-interrogative *vai* IDISJ:**

Olen	onnellinen,	[$_{\text{cond}}$	jos	[$_{\text{JP}}$	Pekka	{	✓*tai*	/	**vai*	}	Liina]	tulee	].
COP.1sg	happy		if		Pekka		SDISJ	/	IDISJ		Liina	comes	

‘I will be happy if Pekka or Liina comes.’ (Hanna Parviainen, p.c.)

Further theoretical study of such interrogative and standard disjunctor pairs across different languages of the world, informed by the study here, may lead to new insights into the extent and shape of cross-linguistic variation in the use of logical connectives for the expression of alternatives.

[35] The Finnish interrogative disjunctor *vai* inside a conditional clause can lead to the formation of an alternative question at the level of the containing clause, but this requires the addition of a question particle *-ko*, making the *vai* option in (75) simply ungrammatical.

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Appendix A Against the clausal disjunct analysis of alternative questions

James Huang, Audrey Li, and Li Yafei (2009: 250–257) and Ray Huang (2009, 2010a,b) both propose that Mandarin alternative questions always involve disjunction of full clauses — described as "full-size, bi-clausal sources" by J. Huang et al. (2009: 250) and of "TP/IP" size by R. Huang (2009) — but that disjuncts can sometimes appear to be subclausal on the surface due to the application of productive pro-drop and ellipsis processes, as well as an operation of so-called Conjunction Reduction. In this Appendix, I argue against these clausal disjunct proposals, in support of my own approach whereby both *háishi* interrogative disjunctions and *huòzhe* standard disjunctions can take disjuncts of variable size.

Concretely, I illustrate how the clausal disjunct approach derives the grammatical example in (76a), repeated from (23b) above without my bracketing. (76b) is a slightly simplified form of the derivation that R. Huang gives for this sentence in R. Huang 2010a: 127.

(76) **Deriving apparent local disjunction via Conjunction Reduction:** (R. Huang, 2010a: 123, 127)

a. Nǐ xǐhuān Zhāng Sān *háishi* Lǐ Sì xiě de shū?
   2sg like Zhang San IDISJ Li Si write DE book
   'Do you like the books that Zhang San wrote or the books that Li Si wrote?'

b. [$_{\text{TP}}$ Nǐ xǐhuān Zhāng Sān ~~xiě de shū~~] *háishi* [$_{\text{TP}}$ *pro* ~~xǐhuān~~ Lǐ Sì xiě de shū]?
   2sg like Zhang San write DE book IDISJ like Li Si write DE book

As the derivation in (76b) makes clear, general processes such as pro-drop (proposed by R. Huang for the second clause's subject) are insufficient to arrive at the surface word order in (76a). These works therefore adopt the use of Conjunction Reduction (Chomsky, 1957; Ross, 1967; Wang, 1967), a non-constituent deletion process proposed to specifically apply within coordinate structures, that will "delete the identical constituent[s] from the edge of conjuncts in coordinate sentences... forward deletion applies where a coordinate structure shows an identical element on a left branch, whereas backward deletion applies the other way around" (R. Huang, 2010a: 98). Here, this results in the forward deletion of *xǐhuān* at the left edge of the right conjunct and backward deletion of *xiě de shū* at the right edge of the left conjunct.

Maintaining the view that Mandarin alternative questions uniformly involve disjuncts of clausal size thus commits us to the existence of some specialized non-constituent deletion operation such as Conjunction Reduction, the precise nature and properties of which are left unclear, and which has been abandoned in most subsequent work (see e.g. Gazdar 1981, although see also Schein 2017). This approach also faces an overgeneration problem. As illustrated in (76), Conjunction Reduction as invoked here must be insensitive to syntactic barriers such as clause boundaries or islands, thereby accounting for the apparently island-insensitive nature of *háishi* interrogative disjunctions, as reviewed in section 3.1 above. At the same time, as Erlewine (2014) notes, this structural insensitivity fails to predict the sensitivity to intervention effects or *wh*-island effects, an instance of which is in (78) below. (I discuss *wh*-island effects further in Appendix B.) Concretely, for example, Conjunction Reduction as in (76) predicts no difference between the (a) and (b) examples in (77), repeated from above, and (78), contrary to fact. See also J. Huang 1988a: 686 and He 2011: 89 for additional examples that are challenging for a Conjunction Reduction account.

(77) **Focus intervention effect unexplained by Conjunction Reduction:** =(30a, 31)

a. * **Zhǐyǒu** [Zhāng Sān]$_F$ chī-le píngguǒ *háishi* júzi (ne)?
only Zhang San eat-PFV apple IDISJ orange NE
Intended: 'Was it an apple or an orange that only Zhang San ate?'

b. [Zhǐyǒu [Zhāng Sān]$_F$ chī-le píngguǒ (ne)] *háishi* [zhǐyǒu [Zhāng Sān]$_F$ chī-le júzi (ne)]?
only Zhang San eat-PFV apple NE IDISJ only Zhang San eat-PFV orange NE
'Did only Zhang San eat an apple or did only Zhang San eat an orange?'

(78) ***Wh*-island effect unexplained by Conjunction Reduction:**

a. * Nǐ xiǎng zhīdào [*shéi* xǐhuān Lǐ Sì *háishi* Wáng Wǔ] (ne)? (Erlewine, 2014: 226)
2sg want know who like Li Si IDISJ Wang Wu NE
Intended: 'Is it Li Si or Wang Wu that you wonder who likes?'

b. [Nǐ xiǎng zhīdào [*shéi* xǐhuān Lǐ Sì] (ne)] *háishi* [nǐ xiǎng zhīdào [*shéi* xǐhuān Wáng Wǔ] (ne)]?
2sg want know who like Li Si NE IDISJ 2sg want know who like Wang Wu NE
'Do you wonder who likes Li Si or do you wonder who likes Wang Wu?'

Finally, I present additional evidence for the variable size of disjuncts in alternative questions from the distribution of sentence-final *ne* in alternative questions. The sentence-final particle *ne* frequently appears on matrix alternative questions, as it also frequently does with matrix *wh*-questions (see e.g. Cheng, 1991). It is unavailable in embedded questions.[36]

[36] There is also a distinct *ne* that Chao (1968: 805) and others have described as expressing a "continued state" aspectual semantics. See Constant 2011 and 2014: 406–436 for discussion of aspectual *ne* in relation to the *ne* here, with examples demonstrating their distinct semantics, syntactic positions, and embeddability. See also Pan 2019a: 19–21 for related discussion of dialectal variation in the use of aspectual *ne*.

What is of particular importance here is that *ne* can appear multiply when full clauses are disjoined. See example (79) below, as well as examples (77b) and (78b) above. In such cases, with two disjuncts of clausal size, the first, second, or both *ne* can be pronounced at once. Syntactically, Constant (2014), Paul (2014), and Pan (2019a, 2022) (also in Pan and Paul 2016, 2017) have argued that this sentence-final particle *ne* is a head in the CP domain. This indicates that in all such examples where *ne* can appear multiply, the individual disjuncts are of CP size.

(79) **Two *ne* in alternative question with clausal disjuncts:** (Li et al., 1984: 76)

[$_{\text{CP}}$ Zhāng Sān qù ne] *háishi* [$_{\text{CP}}$ Lǐ Sì lái ne]?
Zhang San go NE IDISJ Li Si come NE
'Will Zhang San go or will Li Si come?'

Such structures are compatible with my semantic proposal, as I treat the semantic operator Q (12) as adjoining at the clause edge, rather than being a particular C head itself. The J head realizing *háishi* disjoins the two CPs, forming the set of these propositions as its alternative semantic value; Q adjoins above this entire JP, resulting in the intended alternative question meaning. I abstract away from the semantic contribution of *ne* here, but see Constant 2014, Ito 2023, and citations there for various views.

In contrast, as Constant (2014) notes, *ne* cannot appear at the right edge of disjuncts of sub-clausal size. See for example (80), two PPs are disjoined by *háishi*. Only one *ne* can be pronounced in (80), at the end of the entire utterance. This is exactly what is predicted on my account, where *háishi* interrogative disjunction can take disjuncts of variable size, together with the idea that the particle *ne* is in the CP domain.

(80) ***Ne* cannot be added after sub-clausal disjuncts:** (Constant, 2014: 341)

Tā xiǎng [[$_{\text{PP}}$ gēn Xiǎo-Wáng] (*ne) *háishi* [$_{\text{PP}}$ gēn Xiǎo-Lǐ] (*ne)] jiéhūn (✓ne)?
3sg want with little-Wang NE IDISJ with little-Li NE marry NE
'Does he/she want to marry Wang or Li?' (alternative question)

Constant (2014) however goes on to note that there are also examples such as (81) below, reproduced here without bracketing, which at first glance gives the impression that *ne* may appear at the edge of each disjunct of *háishi*, regardless of their size. I propose that the availability of two *ne* in (81) indicates that, in this case, *háishi* takes two disjuncts of CP size, but that the second disjunct undergoes a form of ellipsis that we might call *stripping* or *bare argument ellipsis* (see e.g. Hankamer and Sag, 1976; Rooth, 1992).

I assume a movement-and-deletion derivation for stripping, as illustrated for (81) in (82) below (see also Merchant, 2003; Wurmbrand, 2017).

(81) **Two *ne* in alternative question with apparent sub-clausal disjuncts:** (Constant, 2014: 341)

Tā xiǎng qǔ Xiǎo-Wáng ne *háishi* Xiǎo-Lǐ ne?
3sg want marry little-Wang NE IDISJ little-Li NE
'Does he/she want to marry Wang or Li?' (alternative question)

(82) [[$_{\text{CP}}$ Tā xiǎng qǔ Xiǎo-Wáng ne] *háishi* [$_{\text{CP}}$ Xiǎo-Lǐ ~~[$_{\text{TP}}$ tā xiǎng qǔ *t*]~~ ne]]
3sg want marry little-Wang NE IDISJ little-Li 3sg want marry NE

Finally, I note that this form of stripping must be forwards deletion rather than backwards; in particular, it is not possible to generate forms such as (83):

(83) **Ungrammatical result of backwards stripping:** (Wu, 1992: 81)

*[[$_{\text{CP}}$ Nǐ Δ ne] *háishi* [$_{\text{CP}}$ wǒ qù Běijīng ne]]?
2sg NE IDISJ 1sg go Beijing NE
literally ≈ 'You, or will I go to Beijing?'

Note that my stripping account does not generally require the correlate of stripping to be clause-final as it is in (81). Stripping with a clause-medial correlate predicts the availability of apparently discontinuous disjunction. Such examples are indeed possible; see (84), where Δ indicates the ellipsis site in the right disjunct. The availability of *ne* in two positions again supports the underlying presence of two CPs in this case.

(84) **Stripping with a clause-medial correlate:**

[[$_{\text{CP}}$ Tā xiǎng [$_{\text{PP}}$ gēn Xiǎo-Wáng] jiéhūn (ne)] *háishi* [$_{\text{CP}}$ [$_{\text{PP}}$ gēn Xiǎo-Lǐ] Δ (ne)]]?
3sg want with little-Wang marry NE IDISJ with little-Li NE
'Does he/she want to marry Wang or Li?' (alternative question)

Although Constant introduces the data in (80) and (81), he does not offer an analysis, concluding that "the syntactic restrictions remain to be explained" (p. 342). My account here offers a natural explanation for such data. In (81), the correlate *Xiǎo-Wáng* of the stripping constituent *Xiǎo-Lǐ* is clause-final, giving the illusion of a possible local disjunction parse, but it is actually the disjunction of two full clauses with stripping.

It is important to note that, although I invoke the possibility of clausal disjuncts with ellipsis in (81–82) here, my proposal differs from that of J. Huang et al. (2009) and R. Huang (2009, 2010a,b) in two ways. First, the form of ellipsis I invoked is the well established and independently necessary form of stripping, which as sketched in (82) above can be derived by constituent (TP) ellipsis. This is contrast to the prior accounts which resort to a non-constituent Conjunction Reduction operation. Second, on my account *háishi* can take full clause disjuncts but can also take disjuncts of sub-clausal size. No clausal disjunction parse is possible in (80), explaining the unavailability of *ne* following its disjuncts, making such examples a strong argument that *háishi* interrogative disjunction may take disjuncts of variable size.

Appendix B Interactions between disjunctions and *wh*-phrases

In section 3.1, I noted that Erlewine (2014) reports that *háishi* interrogative disjunction cannot scope out of an embedded *wh*-question, i.e. that it is subject to *wh*-island effects. Consider example (85) below. Note that the embedding *xiǎng zhīdào* must embed a question. Erlewine (2014) only reports the unavailability of reading (a), but it is worth noting that the utterance is also ungrammatical with the intended reading in (b). I also comment below on the unavailability of embedded multiple question parses.

(85) ***wonder* [... *wh* ... IDISJ ...]** (based on Erlewine, 2014: 234)

*Zhāng Sān xiǎng zhīdào [*shéi* xǐhuān [$_{JP}$ Lǐ Sì *háishi* Wáng Wǔ]] (ne)?
Zhang San want know who like Li Si IDISJ Wang Wu NE

a. * 'Is it LS or WW *y* that ZS wonders who likes *y*?' (matrix alt. q, embedded *wh* q.)

b. * 'Who *x* is it that ZS wonders whether *x* likes LS or WW?' (matrix *wh* q., emb. alt. q.)

R. Huang (2020: 234) corroborates Erlewine's observation, also presenting additional examples of the same form (although see also Dong 2009: 19 for a potential counterexample). Interestingly, however, all of these examples in the literature have the *wh*-phrase preceding the *háishi* interrogative disjunction within the embedded clause. Where the two are reversed, the utterance *does* have a felicitous use as a matrix alternative question, as in (86a).

(86) ***wonder* [... IDISJ ... *wh* ...]**

Zhāng Sān xiǎng zhīdào [[JP Lǐ Sì *háishi* Wáng Wǔ] xǐhuān *shéi*] (ne)?
Zhang San want know Li Si IDISJ Wang Wu like who NE

a. ✓'Is it LS or WW *y* that ZS wonders who *y* likes?' (matrix alt. q, embedded *wh* q.)

b. * 'Who *x* is it that ZS wonders whether LS or WW likes *x*?' (matrix *wh* q., emb. alt. q.)

Concretely, adopting the distinguished variable approach to alternative computation (Kratzer, 1991; Wold, 1996; Beck, 2016; Howell et al., 2022) which allows for selective binding of alternative sources, the attested reading in (86a) can be computed by the embedded Q specifically associating with *shéi* 'who,' leaving the alternatives introduced by the disjunction for evaluation by matrix Q. See Howell et al. 2022 for arguments that Q is selective in this way, unlike focus-sensitive operators, explaining cross-linguistic asymmetries between insensitivity to *wh*-islands and sensitivity to focus intervention effects. However, this approach leaves the unavailability of the readings in (85a,b) and (86b) surprising. I will leave this puzzle here for future work.

What is of further interest for the subject of this paper is the behavior of *huòzhe* standard disjunction in these same contexts. Specifically, we observe in (87–88) below that the ability of *huòzhe* to scope out of the embedded question, in the (a) readings, appears to parallel the availability of the matrix alternative questions in (85–86) above.

(87) ***wonder* [... *wh* ... SDISJ ...]**

Zhāng Sān xiǎng zhīdào [*shéi* xǐhuān [JP Lǐ Sì *huòzhe* Wáng Wǔ]].
Zhang San want know who like Li Si SDISJ Wang Wu

a. * 'For either LS or WW *y*, ZS wonders who likes *y*.' (wide scope disjunction)

b. ✓'ZS wonders who likes either one of LS or WW.' (narrow scope disjunction)

(88) ***wonder* [... SDISJ ... *wh* ...]**

Zhāng Sān xiǎng zhīdào [[JP Lǐ Sì *huòzhe* Wáng Wǔ] xǐhuān *shéi*].
Zhang San want know Li Si SDISJ Wang Wu like who

a. ✓'For either LS or WW *y*, ZS wonders who *y* likes.' (wide scope disjunction)

b. ✓'ZS wonders who either one of LS or WW likes.' (narrow scope disjunction)

In sum, whatever mechanisms underly these contrasts, the fact that they apply in parallel to alternative question formation with *háishi* and the scope-taking of *huòzhe* directly supports my account, where both disjunctors share a syntactic and semantic core that projects their disjuncts as alternatives.

Finally, I also note that *háishi* interrogative disjunctions cannot form a multiple question together with a *wh*-phrase, as noted by Yang (2012: 48, 2015: 156), Duan (2015), Ito (2016: 352), Fu (2020: 497), and R. Huang (2020: 234). Accordingly, examples such as (85–86) above also lack embedded multiple question readings. (See also Szabolcsi 1997, especially note 6 on pages 322–323, for further discussion of disjunctions in and of *wh*-questions.) An anonymous reviewer also notes that a sentence with multiple *háishi* interrogative disjunctions likewise cannot be interpreted as a multiple question. (See also Beck and Kim 2006: 187–188 for discussion of the uncertain status of parallel constructions in English.) I will leave the deeper explanation for this apparent inability to form multiple questions involving interrogative disjunction for future work.